BASE24 RELEASE 6.0

IMNI/IMNSC INSTALLATION GUIDE

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Version 7

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REVISION HISTORY

Record of Changes

Author	Date	Version	Comments
ACI	January 15,2004	A	Initial Release.
ACI	April 26,2004	1.0	Update the LNCF description.
ACI	June 30, 2004	2.0	Added EMV support
ACI	October 28, 2004	3.0	Added TIM support
ACI	November 10, 2004	3.0	Added CDNMACS make stream references
ACI	September 22, 2006	3.0	Added LNCF parameter to suppress Track 2 discretionary data for Credit card Cash Advance from ATM.
ACI	April 2, 2007	4.0	Set TSS flag to true
ACI	November 21, 2007	5.0	ICF Configuration updates for IMNSC Switch Added Appendix B
ACI	October 2009	6.0	Expand Bin Table
ACI	November 2011	7.0	Binding under BASE24 R6.V10.

Change Control Index

Change Control Reference	Description of Change	Modules Effected

Related Documents

Document Name	Author
BASE24-CLASSIC 6.0 Install Guide	ACI
BASE24-Transaction Security Manual	ACI
BASE24-Transaction Security Services Processing Guide	ACI
BASE24 Release 6.0 Canadian Users Conversion Implementation Guide	ACI
Interac EMV support in IMNsc Interface External Specification	ACI
BASE24-pos IMNi Switch Interface Log Messages Manual – v1.7	ACI

BASE24-pos IMNsc Switch Interface Log Messages Manual – v3.1	ACI
IDP Functional Specifications	Interac
SCD Functional Specifications	Interac

TABLE OF CONTENTS

1. INTRODUCTION

This document describes how to install and configure BASE24 Release 6.0 IMNI and IMNSC switch interface software.

The IMNI and IMNSC switch interfaces are used for switching transactions to the Interac Member Node (IMN) from/to BASE24. The IMNI switch interface is for BASE24-pos systems and the IMNSC switch interface is for BASE24-atm systems. Both IMNI and IMNSC 6.0 are developed under BASE24 6.0 release to support the Transaction Security Service (TSS).

It is essential that the procedures and guidelines described be followed closely and in the proper sequence to ensure a successful implementation. For any HP Nonstop server system configuration problems please refer to the appropriate HP Nonstop server manuals.

If applicable, members must reapply any currently supported custom (i.e. RPQ/CSM) code after the release is installed.

1.1 Intended Audience

The intended audience of this document is system administrators, switch product support personnel or any IT professional with switch product knowledge.

The installer must have working knowledge of the HP Nonstop server (Tandem) environment, the Backup, Restore and MAKE utilities, and the IMNI/IMNSC products. Familiarity with the naming conventions used and the location of relevant BASE24 components for compiling and configuring purposes is also required.

1.2 Getting Started

Before starting, the following information must be prepared:

- A user ID and password that has access to the BASE24 environment.
- A BASE24 user ID and password.
- Location of installed BASE24 product.
- IMN node information to configure BASE24 (i.e. Node ID, Node name).
- Security key information for BASE24.

Please refer to the BASE24 Classic 6.0 Installation Guide for more details.

1.3 Assumptions

- All BASE24 6.0 product files are received and installed on your system before running MAKE.
- BASE24 installation procedures are not included in this document.
- Transaction Security Service (TSS) installation procedures are not included in this document.

2. DOCUMENT SCOPE

This document describes the installation and configuration procedures for the following BASE24 Release 6.0 components:

- IMNI files.
- IMNSC files.
- IMNA & ICF Pathway requesters and servers.
- IMNI, IMNI Manager, IMNSC and IMNSC manager XPNET Satellite process configuration.
- LNCF (formerly LCONF) parameters and assigns.
- Data files.
- ICF records.
- KEY records.
- ISECD.

This document is NOT written for a specific BASE24 customer or particular network environment.

3. RELEASE CONTENTS

3.1 IMNI Files

1. SW60IMNI.IMNACM	:	IMNI Admin Cleanup MAKE
2. SW60IMNI.IMNACS	:	IMNI Admin Cleanup Source
3. SW60IMNI.IMNEXTEM	:	IMNI MAKE for External File Generation
4. SW60IMNI.IMNIARM	:	IMNI Alternate Routing MAKE
5. SW60IMNI.IMNIARS	:	IMNI Alternate Routing Source
6. SW60IMNI.IMNICMDM	:	IMNI NCS Operator Command Processor MAKE
7. SW60IMNI.IMNICMDS	:	IMNI NCS Operator Command Processor Source
8. SW60IMNI.IMNID	:	IMNI EMV Dummy Source file
9. SW60IMNI.IMNIDCLS	:	IMNI Global Data Declarations
10. SW60IMNI.IMNIDDLM	:	IMNI DDL MAKE
11. SW60IMNI.IMNIDDLS	:	IMNI DDL Source
12. SW60IMNI.IMNIEMVD	:	IMNI EMV Dummy Source file
13. SW60IMNI.IMNIFIOM	:	IMNI File I/O MAKE
14. SW60IMNI.IMNIFIOS	:	IMNI File I/O Source
15. SW60IMNI.IMNIINIM	:	IMNI Initialization Procedure MAKE
16. SW60IMNI.IMNIINIS	:	IMNI Initialization Procedure Source
17. SW60IMNI.IMNILFXG	:	ILF Extract Global
18. SW60IMNI.IMNILFXM	:	ILF Extract MAKE
19. SW60IMNI.IMNILFXS	:	ILF Extract Source
20. SW60IMNI.IMNIM	:	IMNI MAKE
21. SW60IMNI.IMNIMLM	:	IMNI Mainline MAKE
22. SW60IMNI.IMNIMLS	:	IMNI Mainline Source
23. SW60IMNI.IMNIOPTS	:	IMNI Compile Options
24. SW60IMNI.IMNIPSTM	:	IMNI PSTM Processor MAKE
25. SW60IMNI.IMNIPSTS	:	IMNI PSTM Processor Source
26. SW60IMNI.IMNISCNS	:	IMNI Screen Source
27. SW60IMNI.IMNISEMM	:	IMNI SEM Processor MAKE
28. SW60IMNI.IMNISEMS	:	IMNI SEM Processor Source
29. SW60IMNI.IMNIUTLM	:	IMNI Utility Procedures MAKE
30. SW60IMNI.IMNIUTLS	:	IMNI Utility Procedures Source
31. SW60IMNI.IMNMCMDM	:	IMNM NCS Operator Command Processor MAKE
32. SW60IMNI.IMNMCMDS	:	IMNM NCS Operator Command Processor Source
33. SW60IMNI.IMNMDCLS	:	IMNM Global Data Declarations
34. SW60IMNI.IMNMINIM	:	IMNM Initialization Procedures MAKE
35. SW60IMNI.IMNMINIS	:	IMNM Initialization Procedures Source
36. SW60IMNI.IMNMM	:	IMNM MAKE
37. SW60IMNI.IMNMMLM	:	IMNM Mainline MAKE
38. SW60IMNI.IMNMMLS	:	IMNM Mainline Source
39. SW60IMNI.IMNMOPTS	:	IMNM Compile Options
40. SW60IMNI.IMNMUTLM	:	IMNM Utility Procedures MAKE
41. SW60IMNI.IMNMUTLS	:	IMNM Utility Procedures Source
42. SW60IMNI.MAKEIMNI	:	IMNI Local MAKE File
43. SW60IMNI.RQCGFM	:	CGF Requester MAKE
44. SW60IMNI.RQCGFS	:	CGF Requester Source

45. SW60IMNI.RQIMNAM : IMNI Admin Requester MAKE
46. SW60IMNI.RQIMNAS : IMNI Admin Requester Source
47. SW60IMNI.RQIMNIM : ICF-IMN Requester MAKE
48. SW60IMNI.RQIMNIS : ICF-IMN Requester Source
49. SW60IMNI.SCRNCGFS : CGF Screen Source
50. SW60IMNI.SVCGFM : CGF Server MAKE
51. SW60IMNI.SVCGFS : CGF Server Source
52. SW60IMNI.SVIMNAM : IMNI Admin Server MAKE
53. SW60IMNI.SVIMNAS : IMNI Admin Server Source
54. SW60IMNI.SWIMNIFM : IMNI MAKE Configuration
55. SW60IMNI.SWIMNIM : IMNI MAKE File
56. SW60IMNI.SWIMNIMM : IMNI File Macros (Defines)

3.2 IMNi EMV Files

If you have purchased the EMV product, the following files will also be included.

1. SW60IIMN.IIMNFM - IMNi EMV MAKE Configuration
2. SW60IIMN.IIMNMM - IMNi EMV File Macros
3. SW60IIMN.IMNIEMVM - IMNi EMV MAKE
4. SW60IIMN.IMNIEMVS - IMNi EMV Source
5. SW60IIMS.SWIIMNM - IMNi MAKE

3.3 IMNSC Files

1. SW60IMSC.IMNACM : IMNSC Admin Cleanup MAKE
2. SW60IMSC.IMNACS : IMNSC Admin Cleanup Source
3. SW60IMSC.IMNMCMDS : IMNM NCS Operator Command Processor Source
4. SW60IMSC.IMNMDCLS : IMNM Global Data Declarations
5. SW60IMSC.IMNMINIS : IMNM Initialization Procedures Source
6. SW60IMSC.IMNMM : IMNM MAKE
7. SW60IMSC.IMNMMLS : IMNM Mainline Source
8. SW60IMSC.IMNMOPTS : IMNM Compile Options
9. SW60IMSC.IMNMUTLS : IMNM Utility Procedures Source
10. SW60IMSC.IMSCARS : IMNSC Alternate Routing Source
11. SW60IMSC.IMSCCMDS : IMNSC NCS Operator Command Processor Source
12. SW60IMSC.IMSCD : IMNSC EMV Dummy Source file
13. SW60IMSC.IMSCDCLS : IMNSC Global Data Declarations
14. SW60IMSC.IMSCDDLM : IMNSC DDL MAKE
15. SW60IMSC.IMSCDDLS : IMNSC DDL Source
16. SW60IMSC.IMSCEMVD : IMNSC EMV Dummy Source file
17. SW60IMSC.IMSCFIOS : IMNSC File I/O Source
18. SW60IMSC.IMSCINIS : IMNSC Initialization Procedure Source
19. SW60IMSC.IMSCLFXG : ILF Extract Global
20. SW60IMSC.IMSCLFXM : ILF Extract MAKE
21. SW60IMSC.IMSCLFXS : ILF Extract Source
22. SW60IMSC.IMSCM : IMNSC MAKE
23. SW60IMSC.IMSCMLS : IMNSC Mainline Source

24. SW60IMSC.IMSCOPTS : IMNSC Compile Options
25. SW60IMSC.IMSCSCNS : IMNSC Screen Source
26. SW60IMSC.IMSCSEMS : IMNSC SEM Processor Source
27. SW60IMSC.IMSCSTMS : IMNSC STM Processor Source
28. SW60IMSC.IMSCUTLS : IMNSC Utility Procedures Source
29. SW60IMSC.MAKEIMSC : IMNSC Local MAKE File
30. SW60IMSC.RQCGFM : CGF Requester MAKE
31. SW60IMSC.RQCGFS : CGF Requester Source
32. SW60IMSC.RQIMNAM : IMNSC Admin Requester MAKE
33. SW60IMSC.RQIMNAS : IMNSC Admin Requester Source
34. SW60IMSC.RQIMSCM : ICF-IMN Requester MAKE
35. SW60IMSC.RQIMSCS : ICF-IMN Requester Source
36. SW60IMSC.SCRNCGFS : CGF Screen Source
37. SW60IMSC.SVCGFM : CGF Server MAKE
38. SW60IMSC.SVCGFS : CGF Server Source
39. SW60IMSC.SVIMNAM : IMNSC Admin Server MAKE
40. SW60IMSC.SVIMNAS : IMNSC Admin Server Source
41. SW60IMSC.SWIMSCFM : IMNSC MAKE Configuration
42. SW60IMSC.SWIMSCM : IMNSC MAKE File
43. SW60IMSC.SWIMSCMM : IMNSC File Macros (Defines)

3.4 IMNsc EMV Files

If you have purchased the EMV product, the following files will also be included.

1. SW60IIMS.IIMSFM - IMNsc EMV MAKE Configuration
2. SW60IIMS.IIMSMM - IMNsc EMV File Macros
3. SW60IIMS.IMSCEMVM - IMNsc EMV MAKE
4. SW60IIMS.IMSCEMVS - IMNsc EMV Source
5. SW60IIMS.SWIIMSM - IMNSC MAKE

3.5 CDNMACS Macro Definition File

This file contains all of the flags and file definitions for the Canadian Switch software.

1. BA60CDN.CDNMACS - Canadian macro definition file

You will need to update your startup make file (B24C, or any other file you use to run a global make) to point your **=allmacs** define to this **cdnmacs** file:

```
delete define =allmacs
   add    define =allmacs,     file $cdn60.ba60cdn.cdnmacs
 ==add    define =allmacs,     file $b2460.ba60make.custmacs
```

4. TIM ENHANCEMENTS

The IMNi and IMNsc Interchange Interfaces support the Transactional Interface Monitoring (TIM) Enhancement. This enhancement provides the ability to monitor, in real-time, the actual transaction health between BASE24 and the interchange. TIM provides tracking of more than a dozen transaction conditions/counters at the process level. Depending upon the particular interface, this enhancement supports issuer and acquirer traffic for both BASE24-atm and BASE24-pos transactions. This enhancement requires ACI's NET24-OM (XN-NT100) and Compaq's Measure product.

This enhancement impacts the IMNi and IMNsc interfaces by replacing the current "measure" functionality with the TIM "pmon" functionality.

4.1 IMNi Modules Affected

The following IMNi modules were affected by the TIM enhancement:

1. SW60IMNI.IMNICMDM
2. SW60IMNI.IMNICMDS
3. SW60IMNI.IMNIDCLS
4. SW60IMNI.IMNIFIOM
5. SW60IMNI.IMNIFIOS
6. SW60IMNI.IMNIM
7. SW60IMNI.IMNIMLS
8. SW60IMNI.IMNIUTLS
9. SW60IMNI.SWIMNIFM

4.2 IMNsc Modules Affected

The following IMNsc modules were affected by the TIM enhancement:

1. SW60IMSC.IMSCCMDS
2. SW60IMSC.IMSCDCLS
3. SW60IMSC.IMSCFIOS
4. SW60IMSC.IMSCM
5. SW60IMSC.IMSCMLS
6. SW60IMSC.IMSCUTLS
7. SW60IMSC.SWIMSCFM
8. SW60IMSC.SWIMSCM

4.3 BASE Modules Affected

The BA60CDN.CDNMACS file has been updated to allow building of the IMNi and IMNsc interfaces with the TIM enhancement. The following defines were introduced:

swimni_perfmon_on	=	$(true)
swimsc_perfmon_on	=	$(true)
swswpm_swpmonmm	=	$b2460.sw60swpm.swpmonmm

The standard PMON code is needed and will be included using the following SWPMONMM dependency in the CDNMACS file:

```
#-----------------------------------------------------------------
# TIM Enhancement                                                -
#-----------------------------------------------------------------
!IF $(swimni_perfmon_on) = TRUE |THEN|
     !include $(swswpm_swpmonmm)
!ENDIF

!IF $(swimsc_perfmon_on) = TRUE |THEN|
     !include $(swswpm_swpmonmm)
!ENDIF
```

4.4 Log Message Handling

Some log messages have changed due to the TIM enhancement. Please refer to the IMNi and IMNsc log message word documents for details.

5. BASE24 INSTALLATION - GETTING STARTED

This section describes how to restore the IMNI/IMNSC software from the distribution media to your system.

Please review all instructions carefully and plan the installation accordingly. Also, please ensure that all required steps are completed in the order described. Disk space requirements should be considered before proceeding with the installation.

The distribution media supplied includes the software licensed for your site.

5.1 Restore the Release Tape(s)/Cartridge(s)

Restore the files from the distribution media to your system following these steps. The actual placement of the files during the restoration process is discussed in the next section.

1. Logon to the HP Nonstop server with the user ID that owns the BASE24 software.

2. Mount the BASE24 release tape/cartridge onto a tape drive/cartridge drive, making sure the write ring has been removed on the tape or the write tab has been set on the cartridge.

3. At a TACL prompt, enter the following command:

   > RESTORE / out $s.#atape, nowait / $TAPE, (*.*.*), VOL $<vol>, LISTALL, OPEN, MYID, TAPEDATE, AUDITED

   Where $<VOL> represents the disk volume to which you wish to restore your files.

If the files have been delivered on a diskette, restore the files to your hard drive and use IXF, FTP, or similar software to upload the files to your HP Nonstop server in the appropriate location.

5.2 Restore the Release CD

To restore the code contained from the CD, please perform the following:

1. Copy the desired file(s) to your hard drive.

2. Transfer Code 100 file(s) in binary format and transfer code 101 file(s) in ASCII format to a HP Nonstop system.

3. Restore the PAK file code to 100 by using the following command:

   > FUP ALTER <PAK file>, CODE 100

4. Run the self extracting PAK file by using the following command:

   > RUN <PAK file>, *.*.*, LISTALL, MYID, VOL <$vol>

   Where $<VOL> represents the disk volume to which you wish to restore your files.

5.3 Release Contents Verification

The release contents should be verified. The Restore output listing should be compared against the listing provided to ensure the release files have been fully and properly restored to the location specified in the Restore execution. Any discrepancies should be promptly reported to ACI.

5.4 Security

As the files are restored, they should be secured to the proper user ID for file creation and program compilation. This user ID is normally the owner of the BASE24 environment or the owner of the source code management system, where applicable.

6. COMPILATION INSTRUCTIONS

In building a new release system, a certain sequence of activities must be enforced. There are dependencies that dictate the compilation order of certain elements. Using the MAKE utility will ensure that these dependencies are met during compilation.

6.1 Binding IMNI and IMNSC code under BASE24 R6.V10

There are two options to bind IMNI/IMNSc with BASE24 R6.V10 BASE, per users' preference and/or policy:

(a) Complete installation of BASE24 R6.V10
- Install BASE24 R6.V10 and all acquired modules
- Compile and bind the IMNi/IMNsc interface

(b) Complete installation of BASE portion of BASE24 R6.V10
- Install only the BASE modules of BASE24 R6.V10 in a different location
- Compile and bind IMNi/IMNsc interface using the BASE modules

Notes:
1) Each of the above options has pros and cons. Users would take the overall strategic views and policy into consideration when making the decision.
   Officially, Option A - a complete upgrade to BASE24 R6.V10 is recommended by ACI.

2) Wherever possible, enable highpin and highrequesters on the newly bound objects.

6.2 MAKE IMNI object files

Before you run the global MAKE, it is necessary to perform the following steps.

- Set the IMNI_ON flag in your CUSTMACS to "true". Please ensure the related product flag is set properly i.e. CLASSIC_ON, POS_ON. For more details on the product flag set up, please refer to the BASE24-CLASSIC 6.0 Install Guide.
- If you have purchased the EMV modules, set the IMNI_EMV_ON flag in your CDNMACS to "true".
- After executing MAKESCAN, please verify the SWIMNIMM and SWIMNIFM to ensure that all the MAKE definitions are pointing to the correct locations. If you have purchased the IMNi EMV modules also verify IIMNFM and IIMNMM.
- Run the global MAKE using the obey file B24C. Change the **=allmacs** define to point to the CDNMACS file instead of the standard CUSTMACS file.
- Verify all the objects are created.
- Set the **tss_key_idx_on** flag to TRUE in CUSTMACS (needed for TSS).

6.3 MAKE IMNSC object files

Before you run the global MAKE, it is necessary to perform the following steps.

- Set the IMNSC_ON flag in your CUSTMACS to true. Please ensure the related product flag is set properly i.e. CLASSIC_ON, ATM_ON. For more details on the product flag set up, please refer to the BASE24-CLASSIC 6.0 Install Guide.
- If you have purchased the EMV modules, set the IMSC_EMV_ON flag in your CUSTMACS to "true".
- After executing MAKESCAN, please verify the SWIMNSCMM and SWIMNSCFM to ensure that all the MAKE definitions are pointing to the correct locations. Any other "M" files to check for EMV If you have purchased the IMNsc EMV modules also verify IIMSFM and IIMSMM.
- Run the global MAKE using the obey file B24C. Change the **=allmacs** define to point to the CDNMACS file instead of the standard CUSTMACS file.
- Verify all the objects are created.
- Set the **tss_key_idx_on** flag to TRUE in CUSTMACS (needed for TSS).

7. IMNI CONFIGURATION

7.1 Create template and data files

1. Use IMNIFUP in SW60IMNI to create the data files as follows:
   - Locate your data subvolume.
   - From TACL, FUP /in SW60IMNI.IMNIFUP/

2. Create an IMNSAF template file by performing the following steps:
   - Locate your template subvolume
   - In FUP,
     - SET LIKE xxxxDATA.IMNSAF
     - CREATE IMNSAF
     - EXIT

   Where xxxx is your network naming convention i.e. TEST.

3. Please ensure the LNCF assign template file locations match the locations for your template files.

4. Using a generic ILF file on the template subvolume, create an ILF template and alternate key for each IMNI process. Please ensure the ILF file name matches the ILF LNCF assign. ACI recommends that the ILF files have "BUFFERED" set. For example:
   - In FUP,
     - SET LIKE ILF
     - SET BUFFERED
     - SET ALTFILE (0,I0YYMMDD)
     - CREATE ILYYMMDD
     - EXIT

Please refer to BASE24 Classic 6.0 Installation Guide to create BASE24-POS template and database files.

7.2 Modify Pathway Configuration File (PATHCONF)

The IMN Admin server must be added to support Admin transactions. If the IMN server is not already in the PATHCONF, please follow these steps:

```
[ IMNA ADMIN SERVER                                    ]
RESET SERVER
SET SERVER CPUS (0:1)
SET SERVER DELETEDELAY 3 HRS
SET SERVER NUMSTATIC 0
SET SERVER PRI 135
SET HOMETERM $SUDO
SET SERVER PROGRAM $volume.xxxxOBJ.SVIMNA
ADD SERVER SERVER-IMNA
```

Where xxxx is your subvolume naming convention i.e. SW60

Please be aware that the IMNI and IMNSC have the same program name for the IMN Admin server and requester pair as well as IMN-ICF requester (T6520-ICF-IMN). Ensure the server and requester are pointing to the appropriate object.

Once the PATHCONF has been changed, you can bring up the PATHWAY system by using the PATHCOLD obey file.

7.3 Add XPNET Processes

You need to add one IMNM and the IMNI processes to your XPNET network.
Using NCPCOM, add the following processes to your XPNET network:

1. The IMNM Process
   - Point the object to SW60IMNI.IMNM.
   - Only one IMNM process can exist in each logical network.
   - Set the priority higher than the IMNI processes by at least 1.

2. IMNI Processes
   - Point the object to SW60IMNI.IMNI.
   - Multiple copies of the IMNI process can be configured in each logical network.
   - Ensure the IMNI symbolic process names exactly match the LNCF parameter (SW-IMNI-PROC-NAMEnn) value.
   - The maximum number of IMNI processes in a logical network is 50.
   - Set the priority of the IMNI processes lower than the IMNM.

7.4 Add IMNI LNCF Assigns & Parameters

Use the LNCF screen to add the following IMNI parameters and assigns from the table below. Default values are given. Changes may be required to suit your environment.

7.4.1 IMNI LNCF Assigns

The following assigns are mandatory and must be set to the appropriate values in your environment.

Assign Name	Assign Value	Description
CGF	\<sys>.$<vol>.<subvol>.IMNCGF	This assign specifies the filename of the Connection Group File (CGF), which the Acquirer IMN Switch Interface uses to route transactions to an issuer node. This assign is required if Automatic Routing is enabled (i.e. IMN-AUTO-ROUTING set to "Enabled") and configured. Otherwise, this assign is not required.
EPCF	\<sys>.$<vol>.<subvol>.EPCF	This assign specifies the filename of the Entry Point Configuration File (EPCF), which contains the Entry Point (EP) information for connecting to the IMN Switch. This file is a non-BASE24 file maintained by the IMN node EPCNTL and EPDEFN screens.
ICF	\<sys>.$<vol>.<subvol>.ICF	This assign specifies the filename of the Interchange Configuration File (ICF), where the IMN Switch

Assign Name	Assign Value	Description
		Interface finds its configuration information. Configuration information includes IMN node ID, switch date, timer and SAF file information.
ILF	\<sys>.$<vol>.<subvolDATA>.ILYYMMDD \<sys>.$<vol>.<subvolTPLT>.ILYYMMDD	This assign specifies the filename of the Interchange Log File (ILF), where the IMN Switch Interface logs transactions that are sent to the switch and received from the switch. Specify both the data location of the file and the template file location. NOTE: Each IMN Switch Interface Process is assigned a unique ILF file and template location.
IMN-ADMN	\<sys>.$<vol>.<subvol>.IMNADMN	This assign specifies the filename of the IMN-ADMN file that contains IMN administrative transactions that are entered by an operator at the IMNA screen.
IMN-BIN	\<sys>.$<vol>.<subvol>.IMNBIN	This assign specifies the filename of the IMN-BIN file that contains card prefixes that the IMN Switch Interface uses to route not-on-us transactions to the appropriate card issuer. This file is a non-BASE24 file maintained by the IMN node.
IMN-IMNM	\<sys>.$<process_id>	This assign specifies the process name of the IMNM process.
IMN-SAF-TEMPLATE	\<sys>.$<vol>.<subvolDATA>.IMNSAF \<sys>.$<vol>.<subvolTPLT>.IMNSAF	This assign specifies the filename of the IMN Switch Interchange process SAF file. The IMNM process uses the template file location to create the SAF file for the IMN Switch Interchange process.
KEYF	\<sys>.$<vol>.<subvol>.KEYF	This assign specifies the filename of the KEY File, which contains the key management information.

Assign Name	Assign Value	Description
LMCF	\<sys>.$<vol>.<subvol>.LMCF	This assign specifies the filename of the Log Message Configuration File (LMCF), which contains log message information. This is a BASE24 product file.
PATHMON-NAME	$<process_id>	This assigns specifies the BASE24 Pathway process name used by IMN Admin server.

7.4.2 IMNI LNCF Parameters

If the parameter does not have an initial value, you are required to set it to an appropriate value. If you add a parameter and assign a new value to the parameter, the new value will be taken. Otherwise, the default value will be used.

Parameter Name	Parameter Default Value	Description
EVALUATE-SWITCH-DATE	A	Indicates what action the IMN Switch Interface should take if the switch settlement date on the ICF differs from the expected switch settlement date. The expected switch date is calculated using the current system date and the IMN cutover times. Valid values are: "A" – Abend. "B" – Use ICF switch date even if different from the expected date and log a warning. "E" – Use expected date and log a warning.
IMN-AUTO-ROUTING	DISABLED	This parameter enables or disables the Auto Routing feature.
IMN-BAL-AND-CUTOVER-START		Indicates the start time of the daily cutover period for the IMN network. This is used to check the transaction capture date. Must be a valid time value using a 24-hour format (HH:MM). Recommended time: 21:00 (Interac mandates cutover window 21:00-21:30).
IMN-BAL-AND-CUTOVER-END		Indicates the end time of the daily cutover period for the IMN network. This is used to check the transaction capture date. Must be a valid time value using a 24-hour format (HH:MM). Recommended time: 21:30 (Interac mandates cutover window 21:00-21:30).

Parameter Name	Parameter Default Value	Description
IMN-DAILY-KEYCHANGE-TIME		Indicates the time of the day to perform daily keychange. Interac requires all working keys to be exchanged automatically once a day. Must be a valid time value using a 24-hour format (HH:MM). Recommended time: a non-busy time, e.g. 03:00 (AM)
IMN-FIL-ID		Indicates the acquirer AIL (FIL) ID of the BASE24 resource node. It is used by the IMN Switch Interchange Process to find a valid entry point in the EPCF. Valid values are "1" to "9".
IMN-HANDSHAKE-TIMEOUT	60	Indicates the amount of time (in seconds) to wait for a handshake response message from the IMN Commonitor process.
IMN-ILF-NOTIFY-INTERVAL	50	Indicates the number of writes to the ILF between each file full percentage check. Min=10, Max =32766.
IMN-ILF-NOTIFY-PERCENTAGE	70	The threshold at which the operator is notified of the ILF's size as a percentage. Min=10, Max=99
IMN-KEYCHANGE-DELAY	5	Indicates the interval (in seconds) between changing keys to a destination and the next. The delay avoids simultaneous keychanges of all keys (MAC, PIN and MSG) to all destinations. The delay is used during startup and daily keychange time.
IMN-MIN-PACING-STAGGER-THRESH	5	Indicates the rate at which pacing messages are sent. This avoids sending pacing messages all at once.
IMN-NOTIFY-COUNTER	7	This parameter specifies a percentage of the key usage counter limit, which is used by the IMN Switch Interface to notify the IMNM of a key change.

Parameter Name	Parameter Default Value	Description
IMN-PACING-ABNORMAL-SECS	30	Indicates the interval (in seconds) between pacing messages sent out by the IMN Switch Interchange Process to the Commonitor when the state of a connection is blocked. Min=10, max=32766. Suggested value: 60
IMN-PACING-NORMAL-SECS	90	Indicates the interval (in seconds) between pacing messages sent out by the IMN Switch Interchange Process to the Commonitor when the state of a connection is normal. Min=10, max=32766. Suggested value: 120
IMN-PACING-RECENT-WEIGHTING	70	Indicates how recent the weighting is that is used to calculate the average response time from the POS product. A value of 100 indicates current response time with no consideration of past performance. Valid values are 0 to 100.
IMN-PACING-TIMER-SECS	30	The interval time (in seconds) that the IMN Switch Interface process will use to re-evaluate POS response times for pacing changes. The two IMN pacing parameters work in the following manner: On every transaction sent successfully to an entry point, the IMN Switch Interface process increments a throughput counter. On every response from BASE24, the IMN Switch Interface process keeps track of the response times. Every N seconds the IMNI will examine these throughput counters and the average BASE24 response time over the period. The N seconds is the IMN-PACING-TIMER-SECS parameter. Min=10, max=32766.

Parameter Name	Parameter Default Value	Description
IMN-POS-RESP-TIME-BLOCKED-10TH	100	Indicates the response time (in tenths of seconds) at which the IMN Switch Interface process will declare the POS product blocked and will notify the IMN node of this condition. Min=10, Max=32766
IMN-POS-RESP-TIME-CONGESTED-10TH	500	Indicates the response time (in tenths of seconds) at which the IMN Switch Interface process will declare the POS product congested and will notify the IMN node of this condition. Min=10,max=32766
IMN-PRODUCTION-TEST	TEST	Indicates the operating mode of the IMN Switch Interface process (production or test). The interface process sets its message classes accordingly (e.g. "01" versus "02"). Although valid values are either "Prod"uction or "Test", only "PROD" is currently supported.
IMN-SAF-DEST-FORWARD-PERCENTAGE	50	Indicates the percentage of the SAF file size for a particular destination at which the SAF mode changes to forward-only mode.
IMN-SAF-DEST-NORMAL-PERCENTAGE	30	Indicates the percentage of the SAF file size for a particular destination at which the SAF mode changes to normal mode.
IMN-SAF-DEST-NOTIFY-PERCENTAGE	40	Indicates the percentage of the SAF file size for a particular destination when a warning message is issued.
IMN-SAF-FAST-THRESHOLD	2	Indicates the number of consecutive transactions that meet the criteria of switching to the SAF fast mode.
IMN-SAF-FORWARD-PERCENTAGE	80	Indicates the percentage of the SAF file size for the whole system at which the SAF mode changes to forward-only mode.
IMN-SAF-NORMAL-PERCENTAGE	60	Indicates the percentage of the SAF file size for the whole system at which the SAF mode changes to normal mode.
IMN-SAF-NOTIFY-INTERVAL	50	Indicates the interval (number of SAF writes) that the IMNI's check the size

Parameter Name	Parameter Default Value	Description
		of the SAF file.
IMN-SAF-NOTIFY-PERCENTAGE	70	The threshold, as a percentage of the SAF file size, at which the operator is notified of the SAF file size.
IMN-SAF-RETRANSMIT-INTERVAL	30	Indicates the interval between SAF messages that are resent to a particular destination. Used when the SAF messages have not been acknowledged.
IMN-SAF-SLOW-INTERVAL	60	Indicates the interval between SAF messages sent to any destination and the destination is not responding successfully to the SAF messages.
IMN-SAF-SLOW-THRESHOLD	3	Indicates the number of consecutive transactions that meet the criteria of switching into the SAF slow mode
IMN-TRANSLATE-ACCOUNT-BALANCE	F	This parameter determines how the IMN Switch Interchange process converts the account balance field from CBC (Cipher Block Chaining) to ECB (Electronic Code Book). Valid value is "T" – Translate Interac CBC to ECB for BASE24. "F" – Leave as CBC for BASE24.
INTERNAL-BIN-TABLE-ENTRIES	32000	Indicates the maximum number of entries allowed in the internal BIN table (128000).
INTERNAL-FKT-TABLE-ENTRIES	100	Indicates the maximum number of entries allowed in the AIL (FIL) Key Table internal table.
INTERNAL-DOWN-VERSION-LEVEL	0	Indicates the allowable range below the current Version-level set in the IMNI that a COMMONITOR must have to allow communication to occur.
INTERNAL-RECOVERY-TIMER-SECS	60	This parameter determines the timer interval (in seconds) that the IMN Switch Interface uses to check the state of the entry points and the IMNM. Any entry points that are closed are re-started at this time.
INTERNAL-SAF-MAX-TRANS-OUT	10	Indicates the maximum number of SAF transactions that are outstanding (in-flight).

Parameter Name	Parameter Default Value	Description
INTERNAL-TRACE-BITS	"0000000000000000"	Indicates the initial trace bit values for logging operator trace messages. A string of 16 ones & zeroes represents each of the trace bits. A "0" means the trace bit is OFF and a "1" means the trace bit is ON. The first trace bit (bit number 0) is the right-most (last) position in the string.
INTERNAL-UP-VERSION-LEVEL	0	Indicates the allowable range above the current Version-level set in the IMNI that a Commonitor must have to allow communication to occur.
POS-LN-CUTOVER	00:00	The time remaining prior to the logical network cutover for BASE24-pos.
SAF-LOG-THRESHOLD	20	This parameter is the threshold at which the operator is notified of the number of SAF records in the SAF file. See TO-CHECK-SAF.
SW-ILF-FILE-RETENTION	7	The number of days the ILF will remain on the system before being purged by the IMN Switch Interchange process.
SW-ILF-PURGE-ATTEMPT	7	Indicates the number of days the ILF will remain on the system before it is purged by the IMNI process.
SW-IMNI-PROC-NAMEnn		This parameter defines the symbolic name of an IMN Switch Interchange Process that the IMNM process manages. Add one parameter per IMN Switch Interchange process. Note: Ensure the symbolic process name matches the IMNI process name in the XPNET configuration.
TO-CHECK-SAF	60	This parameter is the timer interval that the IMNM uses to check the SAF record count thresholds. The timer interval is specified in **hundredths** of seconds. See also SAF-LOG-THRESHOLD.

Parameter Name	Parameter Default Value	Description
POS-FORWARD-INST-ID		DH/RTR/AUTH places in frwd-inst-id-num field of pstm. Represents routing and transit number for base24. used by DH/RTR/AUTH proces. 11 digit numeric field default = zeroes.

8. IMNSC CONFIGURATION

8.1 Create template and data files

1. Use IMSCFUP in SW60IMSC to create data files as following:
   - Locate your data subvolume.
   - From TACL, FUP /in SW60IMSC.IMSCFUP/

2. Create an IMNSAF template file by performing the following steps:
   - Locate you template subvolume
   - In FUP,
     - SET LIKE xxxxDATA.IMNSAF
     - CREATE IMNSAF
     - EXIT

   Where xxxx is your network naming convention i.e. TEST.

3. Please ensure the LNCF assign template file locations match the locations where you created the template files.

4. Using a generic ILF file on the template subvolume, create an ILF template and alternate key for each IMNSC process. Please ensure the ILF file name matches the ILF LNCF assign. ACI recommends that the ILF filebe BUFFERED. For example:
   - In FUP,
     - SET LIKE ILF
     - SET BUFFERED
     - SET ALTFILE (0,I0YYMMDD)
     - CREATE ILYYMMDD
     - EXIT

Please refer to the BASE24 Classic 6.0 Installation Guide to create BASE24-ATM template and database files.

8.2 Modify Pathway Configuration File (PATHCONF)

The IMN Admin server must be added to support the Admin transactions. If it is not already in the PATHCONF, please add the following:

```
[ IMNA ADMIN SERVER                                   ]
RESET SERVER
SET SERVER CPUS (0:1)
SET SERVER DELETEDELAY 3 HRS
SET SERVER NUMSTATIC 0
SET SERVER PRI 135
SET HOMETERM $SUDO
SET SERVER PROGRAM $volume.xxxxOBJ.SVIMNA
```

ADD SERVER SERVER-IMNA

Where xxxx is your subvolume naming convention i.e. SW60

Please be aware that the IMNI and IMNSC have the same program name for the IMN Admin server and requester pair as well as the IMN-ICF requester (T6520-ICF-IMN). Ensure the server and requester are pointing to the appropriate object.

Once the PATHCONF has been changed, you can bring up the PATHWAY system by using the PATHCOLD obey file.

8.3 Add XPNET Processes

You need to add one IMNM and the IMNSC processes to your XPNET network.
Using NCPCOM, add the following processes to your XPNET network:

1. The IMNM Process
   - Point the object to SW60IMSC.IMNM.
   - Only one IMNM process can exist in each logical network.
   - Set the IMNM priority higher than the IMNSC processes by at least 1.

2. IMNSC Processes
   - Point the object to SW60IMSC.IMSC.
   - You can have multiple IMNSC processes in each logical network.
   - Ensure the IMNSC symbolic process name matches the IMNSC process name in the ICF screen (page 12).
   - The maximum number of IMNSC processes in each logical network is 15.
   - Set the priority of the IMNSC processes lower than the IMNM.

8.4 Add IMNSC LNCF Assigns & Parameters

Use the LNCF screen to add the following IMNSC parameters and assigns from the table below. Default values are provided. Changes may be required to suit your environment.

8.4.1 IMNSC LNCF Assigns

The following assigns are mandatory and they must be set to the appropriate values in your environment.

Assign Name	Assign Value	Description
CGF	\<sys>.$<vol>.<subvol>.IMNCGF	This assign specifies the filename of the Connection Group File (CGF) that the Acquirer IMN Switch Interface uses to route transactions to an issuer node. This assign is required if Automatic Routing is enabled (i.e. IMN-AUTO-ROUTING set to "Enabled") and configured. Otherwise, this assign is not required.
EPCF	\<sys>.$<vol>.<subvol>.EPCF	This assign specifies the filename of the Entry Point Configuration File (EPCF) that contains the Entry Point (EP) information for connecting to the IMN Switch. This file is a non-BASE24 file maintained by the IMN node EPCNTL and EPDEFN screens.
ICF	\<sys>.$<vol>.<subvol>.ICF	This assign specifies the filename of the Interchange Configuration File (ICF), where the IMN Switch Interface finds its configuration information. Configuration information includes IMN node ID, switch date, timer and SAF file information.

Assign Name	Assign Value	Description
ILF	\<sys>.$<vol>.<subvolDATA>.ILYYMMDD \<sys>.$<vol>.<subvolTPLT>.ILYYMMDD	This assign specifies the filename of the Interchange Log File (ILF), where the IMN Switch Interface logs transactions that are sent to the switch and received from the switch. Specify both the data location of the file and the template file location. NOTE: Each IMN Switch Interface Process is assigned a unique ILF file and template location.
IMN-ADMN	\<sys>.$<vol>.<subvol>.IMNADMN	This assign specifies the filename of the IMN-ADMN file that contains IMN administrative transactions that are entered by an operator at the IMNA screen.
IMN-BIN	\<sys>.$<vol>.<subvol>.IMNBIN	This assign specifies the filename of the IMN-BIN file, which contains card prefixes the IMN Switch Interface uses to route not-on-us transactions to the appropriate card issuer. This file is a non-BASE24 file maintained by the IMN node.
IMN-IMNM	\<sys>.$<process_id>	This assign specifies the process name of the IMNM.
IMN-SAF-TEMPLATE	\<sys>.$<vol>.<subvolDATA>.IMNSAF \<sys>.$<vol>.<subvolTPLT>.IMNSAF	This assign specifies the filename of the IMN Switch Interchange process SAF file. The IMNM process uses the template file location to create the SAF file for the IMN Switch Interchange process.
KEYF	\<sys>.$<vol>.<subvol>.KEYF	This assign specifies the filename of the KEY File, which contains the key management information.

Assign Name	Assign Value	Description
LMCF	\<sys>.$<vol>.<subvol>.LMCF	This assign specifies the filename of the Log Message Configuration File (LMCF), which contains log message information. This is a BASE24 product file.
PATHMON-NAME	$<process_id>	This assigns specifies the BASE24 Pathway process name used by the IMN Admin server.

8.4.2 IMNSC LNCF Parameters

If a parameter does not have an initial value, you are required to set it to an appropriate value. If you add a parameter and assign a new value to the parameter, the new value will be taken. Otherwise, the default value will be used.

Parameter Name	Parameter Default Value	Description
ATM-LN-CUTOVER	00:00	The logical network cutover time for BASE24-atm.
EVALUATE-SWITCH-DATE	A	Indicates what action the IMN Switch Interface should take if the switch settlement date on the ICF differs from the expected switch settlement date. The expected switch settlement date is calculated using the current system date and the IMN cutover times. Valid values are: "A" – Abend. "B" – Use ICF switch date even if different from the expected date and log a warning. "E" – Use expected date and log a warning.
IMN-AUTO-ROUTING	DISABLED	This parameter enables or disables the Auto Routing feature.
IMN-BAL-AND-CUTOVER-START		Indicates the start time of the daily cutover period for the IMN network. This is used to check the transaction capture date. Must be a valid time value using a 24-hour format (HH:MM).
IMN-BAL-AND-CUTOVER-END		Indicates the end time of the daily cutover period for the IMN network. This is used to check the transaction capture date. Must be a valid time value using a 24-hour format (HH:MM).

Parameter Name	Parameter Default Value	Description
IMN-DAILY-KEYCHANGE-TIME		Indicates the time of the day to perform daily keychange. Interac requires all working keys to be exchanged automatically once a day. Must be a valid time value using a 24-hour format (HH:MM)
IMN-FIL-ID		Indicates the acquirer AIL (FIL) ID of the BASE24 resource node. It is used by the IMN Switch Interchange Process to find a valid entry point in the EPCF. Valid values are "1" to "9".
IMN-HANDSHAKE-TIMEOUT	60	Indicates the amount of time (in seconds) to wait for a handshake response message from the IMN Commonitor process.
IMN-ILF-NOTIFY-INTERVAL	50	Indicates the number of writes to the ILF between each file full percentage check. Min=10, Max =32766.
IMN-ILF-NOTIFY-PERCENTAGE	70	The threshold at which the operator is notified of the ILF's size as a percentage. Min=10, Max=99
IMN-KEYCHANGE-DELAY	5	Indicates the interval (in seconds) between changing keys to a. The delay avoids simultaneous keychanges of all keys (MAC, PIN and MSG) to all destinations. The delay is used at startup and during the daily keychange.
IMN-MIN-PACING-STAGGER-THRESH	5	Indicates the rate at which pacing messages are sent. This avoids sending pacing messages all at once.
IMN-NOTIFY-COUNTER	7	This parameter specifies a percentage of the key usage counter limit, which is used by the IMN Switch Interface to notify the IMNM of a key change.
IMN-PACING-ABNORMAL-SECS	30	Indicates the interval (in seconds) between pacing messages sent out by the IMN Switch Interchange Process to the Commonitor when the state of a connection is blocked. Min=10, max=32766.

Parameter Name	Parameter Default Value	Description
IMN-PACING-NORMAL-SECS	90	Indicates the interval (in seconds) between pacing messages sent out by the IMN Switch Interchange Process to the Commonitor when the state of a connection is normal. Min=10, max=32766.
IMN-PACING-RECENT-WEIGHTING	70	Indicates how recent the weighting is that is used to calculate the average response time from the POS product. A value of 100 indicates current response time with no consideration of past performance. Valid values are 0 to 100.
IMN-PACING-TIMER-SECS	30	The interval time (in seconds) at which the IMN Switch Interface process will re-evaluate the response times of ATM in looking for any pacing changes. The two IMN pacing parameters work in the following manner: On every transaction sent successfully to an entry point, the IMN Switch Interface process increments a throughput counter. On every response from BASE24, the IMN Switch Interface process keeps track of the response times. Every N seconds the IMNSC will examine these throughput counters and the average BASE24 response time over the period. The N seconds is the IMN-PACING-TIMER-SECS parameter. Min=10, max=32766.
IMN-ATM-RESP-TIME-BLOCKED-10TH	100	Indicates the response time (in tenths of seconds) at which the IMN Switch Interface process will declare the ATM product blocked and will notify the IMN node of this condition. Min=10, Max=32766

Parameter Name	Parameter Default Value	Description
IMN-ATM-RESP-TIME-CONGESTED-10TH	500	Indicates the response time (in tenths of seconds) at which the IMN Switch Interface process will declare the ATM product congested and will notify the IMN node of this condition. Min=10,max=32766
IMN-PRODUCTION-TEST	T	Indicates the operating mode of the IMN Switch Interface process (production or test). The interface process sets its message classes accordingly (e.g. "01" versus "02"). Valid values are either "Prod"uction or "Test".
IMN-SAF-DEST-FORWARD-PERCENTAGE	50	Indicates the percentage of the SAF file size for a particular destination at which the SAF mode changes to forward-only.
IMN-SAF-DEST-NORMAL-PERCENTAGE	30	Indicates the percentage of the SAF file size for a particular destination at which the SAF mode changes to normal.
IMN-SAF-DEST-NOTIFY-PERCENTAGE	40	Indicates the percentage of the SAF file size for a particular destination when a warning message is issued.
IMN-SAF-FAST-THRESHOLD	2	Indicates the number of consecutive transactions that meet the criteria of switching to SAF fast mode.
IMN-SAF-FORWARD-PERCENTAGE	80	Indicates the percentage of the SAF file size for the whole system at which the SAF mode is changed to forward-only.
IMN-SAF-NORMAL-PERCENTAGE	60	Indicates the percentage of the SAF file size for the whole system at which the SAF mode is changed to normal.
IMN-SAF-NOTIFY-INTERVAL	50	Indicates the interval (number of SAF writes) that the IMNSC processes check the size of the SAF file.
IMN-SAF-NOTIFY-PERCENTAGE	70	The threshold, as a percentage of the SAF file size, at which the operator is notified of the SAF file size.

Parameter Name	Parameter Default Value	Description
IMN-SAF-RETRANSMIT-INTERVAL	30	Indicates the interval between SAF messages resent to a particular destination. Used when the SAF message has not been acknowledged.
IMN-SAF-SLOW-INTERVAL	60	Indicates the interval between SAF messages resent to any destination when the destination is not responding successfully to a SAF message.
IMN-SAF-SLOW-THRESHOLD	3	Indicates the number of consecutive transactions that meet the criteria of switching to the SAF slow mode
IMN-TRANSLATE-ACCOUNT-BALANCE	F	This parameter determines how the IMN Switch Interchange process converts the account balance field from CBC (Cipher Block Chaining) to ECB (Electronic Code Book). Valid value is "T" – Translate Interac CBC to ECB for BASE24. "F" – Leave as CBC for BASE24.
INTERNAL-BIN-TABLE-ENTRIES	32000	Indicates the maximum number of entries allowed in the BIN internal table (128000).
INTERNAL-FKT-TABLE-ENTRIES	100	Indicates the maximum number of entries allowed in the AIL (FIL) Key internal table.
INTERNAL-DOWN-VERSION-LEVEL	0	Indicates the allowable range below the current Version-level set in the IMNSC that a Commonitor must have to allow communication to occur.
INTERNAL-RECOVERY-TIMER-SECS	60	This parameter determines the timer interval (in seconds) that the IMN Switch Interface checks the state of entry points and the IMNM. Any entry points that are closed are re-started at this time.
INTERNAL-SAF-MAX-TRANS-OUT	10	Indicates the maximum number of SAF transactions that are outstanding (in-flight).

Parameter Name	Parameter Default Value	Description
INTERNAL-TRACE-BITS	"0000000000000000"	Indicates the initial trace bit values for logging operator trace messages. A string of 16 ones & zeroes represents each of the trace bits. A "0" means the trace bit is OFF and a "1" means the trace bit is ON. The first trace bit (bit number 0) is the right-most (last) position in the string.
INTERNAL-UP-VERSION-LEVEL	0	Indicates the allowable range above the current Version-level set in the IMNSC that a Commonitor must have to allow communication to occur.
SW-ILF-FILE-RETENTION	7	The number of days the ILF will remain on the system before being purged by the IMN Switch Interchange process.
SW-ILF-PURGE-ATTEMPT	7	Indicates the number of purge attempts on older ILFs excluding those that are under the protection of the SW-ILF-FILE-RETENTION value. For example if the SW-ILF-PURGE-ATTEMPT is set at 7 and SW-ILF-FILE-RETENTION is set at 4, IMNsc will retain 4 days of ILFs and begin purging ILFs that are 5 to 11 days old.
IMN-ACQ-DISCR-DATA-SUPPRS	N	This parameter indicates to IMNsc whether Track 2 discretionary data for Credit card Cash Advance Acquirer transactions should be suppressed when logging to ILF
IMN-ISS-DISCR-DATA-SUPPRS	N	This parameter indicates to IMNsc whether Track 2 discretionary data for Credit card Cash Advance Issuer transactions should be suppressed when logging to ILF.

9. BASE24 FILE CONFIGURATION

9.1 Interchange Interface File (ICF)

This section describes how to configure the interchange record for the IMN Switch Interface Process. The ICF consists of four standard segments: BASE, ATM, POS and Switch Dependent.

The following pre-requisites are required before you can add ICF records.

- The BASE24 Pathway is configured with the ICF server and requester.
- The following information is already identified:
  - Your local IMN node number
  - The responder IMN node numbers of the members you are establishing connectivity with.
  - The IMNSC process symbolic name. This is required for IMNSC only since the IMNI process symbolic name will be configured in the LNCF parameter.

The primary key to the IMNI/IMNSC ICF is a combination of the FIID and the interchange process name. Due to the configuration of the IMNI/IMNSC modules, the Interchange FIID will be "IMN " and the process name field will contain the destination node name (6 bytes character) and the message class (2 bytes character), separated by one space. The production varlue of the message class of IMNI will be "01" and IMNSC is "03" respectively. These two fields together represent a destination for the IMNI/IMNSC.

There are two types of ICF records required to configure for the IMNI/IMNSC process:

- Control Record
- Destination Record

9.1.1 Control Record

The Control Record contains information about the local IMN node. There is only one Control Record in the file. The Control Record has "IMN" in the Interchange FIID field and sixteen (16) zeroes in the Process field. An example of a Control Record is shown below.

Screen 1

```
BASE24-BASE  INTERCHANGE CONFIG       ATM8        07/10/25  11:47  01 OF 13
INTERCHANGE FIID: IMN                    PROCESS: 0000000000000000


                     SWITCH TYPE: IMN
         INTERCHANGE LOGICAL NET: IMNC
                  REPORTING NAME: $S.#IMN

                  INSTITUTION ID: CONTROL-NETWORK
                       SWITCH ID: IMN
                       STATION 1:
                       STATION 2:

                        SIC CODE: 0
                   CURRENCY CODE: 124  (CAN)
                DEFAULT TERM NUM:
        DEFAULT ACQUIRER ID NUM: 00000700298
       CUSTOMER BALANCE DISPLAY: 0  (DON'T DISPLAY OR PRINT)

 RECORD LAST CHANGED: 07/10/25  11:35   BY USER: 0255 , 00000255   CHANGE
 *********************************** BASE24 ***********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                       F12-HELP
```

Screen 11

```
BASE24-IMSC  ICF FILE MAINTENANCE     ATM8        07/10/25  11:49  12 OF 14
INTERCHANGE FIID: IMN                    PROCESS: 0000000000000000

                       IMSC PROCESS NAME

    P8A^IMSC


    DESTINATION-NODE NUMBER:  700298
        ACQUIRER CNXN STATE:       3 (UP)
          ISSUER CNXN STATE:       3 (UP)
                 GMT OFFSET:     300 (MINUTES)
               SAF FILENAME:   $CTE8.ATM8DATA.IMNSAF
          SAF FAST INTERVAL:     500 (100th's OF SECONDS)



 *********************************** BASE24 ***********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                             F12-HELP
```

For field descriptions of screens 1 to 11; please refer to the BASE24 Base File Maintenance Manual.

9.1.2 Destination Record

The Destination Record contains information about the remote IMN node. There can be several Destination Records in the file, one for each remote IMN node. The Destination Record has "IMN" in the Interchange FIID field and node name and message class, separated by a space, in the Process field. An example of a Destination Record for IMNI is shown below.

```
BASE24-BASE  INTERCHANGE CONFIG       ATM8        07/10/25  11:52  01 OF 13
INTERCHANGE FIID: IMN                    PROCESS: C90001 03


                     SWITCH TYPE: IMN
         INTERCHANGE LOGICAL NET: IMNC
                  REPORTING NAME: $S.#IMN

                  INSTITUTION ID: 0080
                       SWITCH ID: IMN
                       STATION 1:
                       STATION 2:

                        SIC CODE: 0
                   CURRENCY CODE: 124  (CAN)
                DEFAULT TERM NUM:
         DEFAULT ACQUIRER ID NUM: 00000700298
        CUSTOMER BALANCE DISPLAY: 0  (DON'T DISPLAY OR PRINT)

 RECORD LAST CHANGED: 07/10/25  11:37   BY USER: 0255 , 00000255   ADD
 *********************************** BASE24 ***********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                       F12-HELP
```

9.1.3 ICF screen 12 – ICF IMNI/IMNSC Data

ICF screen 12 contains IMNI/IMNSC processing information. An example of the screen is illustrated below, followed by descriptions of the fields.

```
INTERCHANGE FIID: IMN                    PROCESS: C90001 03

                       IMSC PROCESS NAME

    P8A^IMSC





    DESTINATION-NODE NUMBER:  700298
        ACQUIRER CNXN STATE:       3 (UP)
          ISSUER CNXN STATE:       2 (KEY EXCHANGE)
                 GMT OFFSET:     300 (MINUTES)
               SAF FILENAME:   $CTE8.ATM8DATA.IMNSAF
          SAF FAST INTERVAL:     500 (100th's OF SECONDS)
```

IMNI/IMNSC Process Name

This field contains the IMNSC process symbolic name for a given IMN node ID only.

<u>Note</u> Please be aware that the IMNI process name is no longer supported on this screen. The IMNI process name is defined in the LNCF parameter. Please refer to the section 6.4.2 IMNI LNCF Parameters for more details.

Destination-Node Number

This field contains the Interac Node ID for this destination. For the Control Record, it will contain the local Node ID. For the Destination Record, it will contain your responder Node ID.

Acquirer CNXN State

This field contains the current connection status information. The possible values are:

"0" (Down)	The Issuer AIL is not available for receiving messages from the Acquirer AIL.
"1" (Ready)	The Issuer AIL is initialized, but it has not received key exchanges from the Acquirer AIL.
"3" (Up)	Working keys have been received and normal traffic is flowing (default value).

Issuer CNXN State

This field contains the current connection status information. The possible values are:

"0" (Down)	The Issuer AIL is marked down by the Acquirer AIL.
"2" (Key Exchange)	The Acquirer AIL is attempting to send a set of working keys to the Issuer AIL (default value).
"3" (Up)	The Issuer AIL has received all working keys and transactions can be sent to the IMN node.

GMT Offset

This field contains GMT offset in minutes.

SAF Filename

This field contains the physical location of the SAF file for this destination node/message class.

SAF Fast Interval

This field indicates the SAF fast interval (in 100th seconds) of outbound SAF messages for this destination node/message class.

9.1.4 ICF screen 13 – Key Change Transaction Counters

ICF screen 13 contains the Key change thresholds and counters, followed by descriptions of its fields.

```
BASE24-IMNI  ICF FILE MAINTENANCE     QF60        03/11/13  18:00  13 OF 14
INTERCHANGE FIID: IMN                     PROCESS: ACITST 01

              KEY CHANGE TRANSACTION COUNTERS  KEY STATES:

         PIN ENCRYPTION KEY (KPE):    100      KPE: 0  0 - OK
        MAC ENCRYPTION KEY (KMAC):    100     KMAC: 0  1 - KEY CHANGE PENDING
           MESSAGE ENCRYPTION KEY:    100      KME: 0  2 - KEY CHANGE REQUIRED
                     MAC FAILURES:      5              3 - KEY CHANGE DISABLE
         MAX BAD PIN BLOCK ERRORS:      5                  (KME only)
     MESSAGE ENCRYPTION FAILURES:       5
                   MACING ENABLED:      Y
                MAC FAILURE ALARM:      4
        KEY SYNCHRONIZATION ALARM:      4
               KEY EXCHANGE ALARM:      4
       KEY EXCHANGE TIMEOUT ALARM:      1
     KEY EXCHANGE REJECTION ALARM:      1
          PACING NORMAL INTERVAL:     120  (SECONDS)
         PACING CONGEST INTERVAL:      60  (SECONDS)

 ********************************** BASE24 **********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                             F12-HELP
```

PIN Encryption Key (KPE)

This field indicates the maximum number of transactions that should occur before the Acquirer IMNI/IMNSC initiates a 0800 PIN key change (KPE).

Key States

This field indicates the state of the given keys (KPE, KMAC and KME). The valid values are:

"0"	No key change in progress.
"1"	Key change in progress.
"2"	Key change required.
"3"	Key change disable.

Please be aware that the KME is supported in IMNI only.

MAC Encryption Key (KMAC)

This field indicates the maximum number of transactions that should occur before the Acquirer IMNI/IMNSC initiates a 0800 Message Authentication key change (KMAC).

Message Encryption Key

This field indicates the maximum number of transactions that should occur before the Acquirer IMNI initiates a 0800 Message Encryption key change (KME). The Message Encryption Key is only supported in the IMNI.

MAC Failures

This field indicates the maximum number of consecutive MAC verification failures allowable for an Issuer FI before the Acquirer IMNI/IMNSC should initiate a 0800 KMAC change.

Max Bad PIN Block Errors

This field indicates the maximum number of consecutive PIN encryption failures that can occur at the Issuer FI before the Acquirer IMNI/IMNSC should initiate a 0800 KPE change.

Message Encryption Failures

This field indicates the maximum number of consecutive message encryption failures that can occur at the Issuer FI before the Acquirer IMNI should initiate a 0800 KME change.

MACing Enabled

This field specifies whether MACing will be enabled or disabled. The valid values are:

"Y"	MACing is enabled. (Default value)
"N"	MACing is disabled.

MAC Failure Alarm

This field indicates the number of times a MAC error may occur before a message is sent to the operator logger (acquirer and issuer).

Key Synchronization Alarm

This field indicates the number of times a key synchronization error may occur before a message is sent to the operator logger (acquirer and issuer).

Key Exchange Alarm

This field indicates the number of times a key conveyance error may occur before a message is sent to the operator logger (acquirer and issuer).

Key Exchange Timeout Alarm

This field indicates the number of times a key change timeout error may occur before a message is sent to the operator logger (acquirer).

Key Exchange Rejection Alarm

This field indicates the number of times a key change rejection may occur before a message is sent to the operator logger (acquirer).

Pacing Normal Interval

This field indicates number of seconds between pacing normal messages to the IMN node.

Pacing Congest Interval

This field indicates the number of seconds between pacing abnormal messages to the IMN node.

9.1.4.1.1 ICF Page 14 – Transaction Logging Information.

Select the transactions you want logged to the ILF by entering a **Y** next to the message type. Enter a **N** if you don't want the IMNI/IMNSC to log the transaction to the ILF.

```
BASE24-IMNI  ICF FILE MAINTENANCE     QF60        03/11/13  18:00  14 OF 14
INTERCHANGE FIID: IMN                    PROCESS: ACITST 01

              TRANSACTION LOGGING CONFIGURATION

                 0110   Y          0210     Y
                 0230   Y          0430     Y
                 0432   Y          0610     Y
                 0612   Y          0630     Y
                 0632   Y          0810     Y






 TRANSACTIONS MARKED '*' ARE ALWAYS LOGGED ON OUTBOUND REQUESTS


 *********************************** BASE24 ***********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                             F12-HELP
```

Transaction Logging Configuration

This field determines whether a transaction will be logged to the ILF.

9.2 KEY File (KEYF)

When using the Transaction Security Service (TSS) subsystem, hardware-encrypted key information is maintained in the TSS database. The hardware-encrypted key fields in the BASE24 database are set to a value known as a *key locator* value. The TSS process uses the key locator value as the primary key value to retrieve the actual hardware-encrypted key information from its database.

In BASE24 Release 6.0, the actual keys are no longer stored in the KEYF record. Instead, a reference to the actual key (Key Locator) is entered into the key field. The actual keys are kept in the Interchange Security File (ISECD). Please refer to the BASE24 Transaction Security Manual for more details.

Data in the ISECD is maintained by the Interface Security Configuration window. Also, after running a KEYF file conversion, the remaining data in the KEYF, such as the PIN and MAC encrypt types, PIN block format and all dynamic key information is maintained by the BASE24 KEYF Configuration window on the ACI Desktop user interface. This allows all security data to be maintained from a single user interface.

If you need to add KEYF and ISECD records after conversion, you must add a new KEYF record from the BASE24 KEYF Configuration window before adding a new ISECD record from the Interface Security Configuration window. Please refer to the BASE24 TSS Processing Guide for more details.

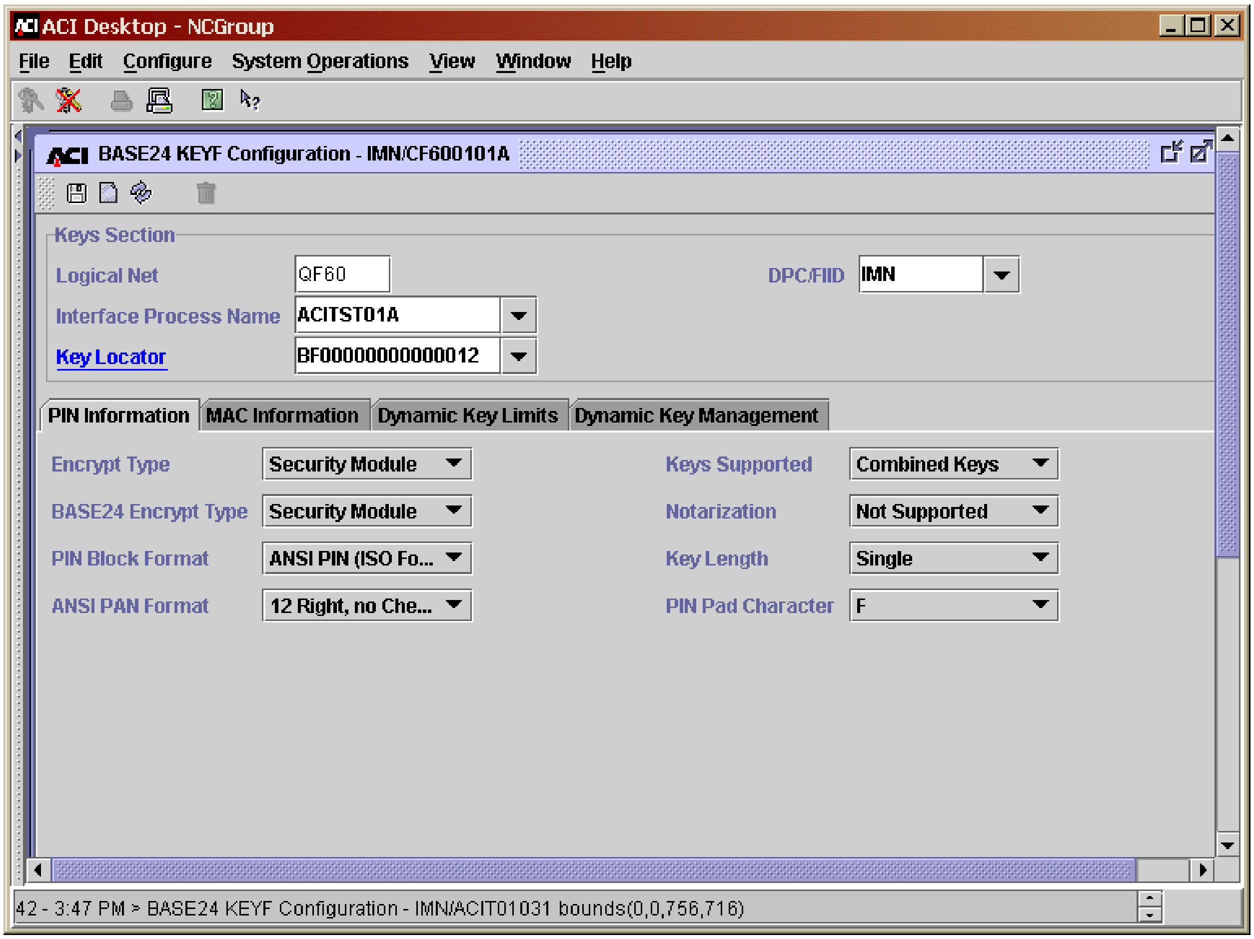
The key in the KEYF record is similar to the key in the ICF record. The DPC/FIID is always "IMN" and the Interface Process name is the concatenation of the node-name, message class and AIL ID ("1" as issuer and "A" as acquirer). In BASE24 6.0 however there are NO spaces in the Interface Process Name field. Basically, all the Destination Records (remote node) in the ICF must be added in the KEYF to set up all of the key information.

9.2.1 KEYF File Conversion

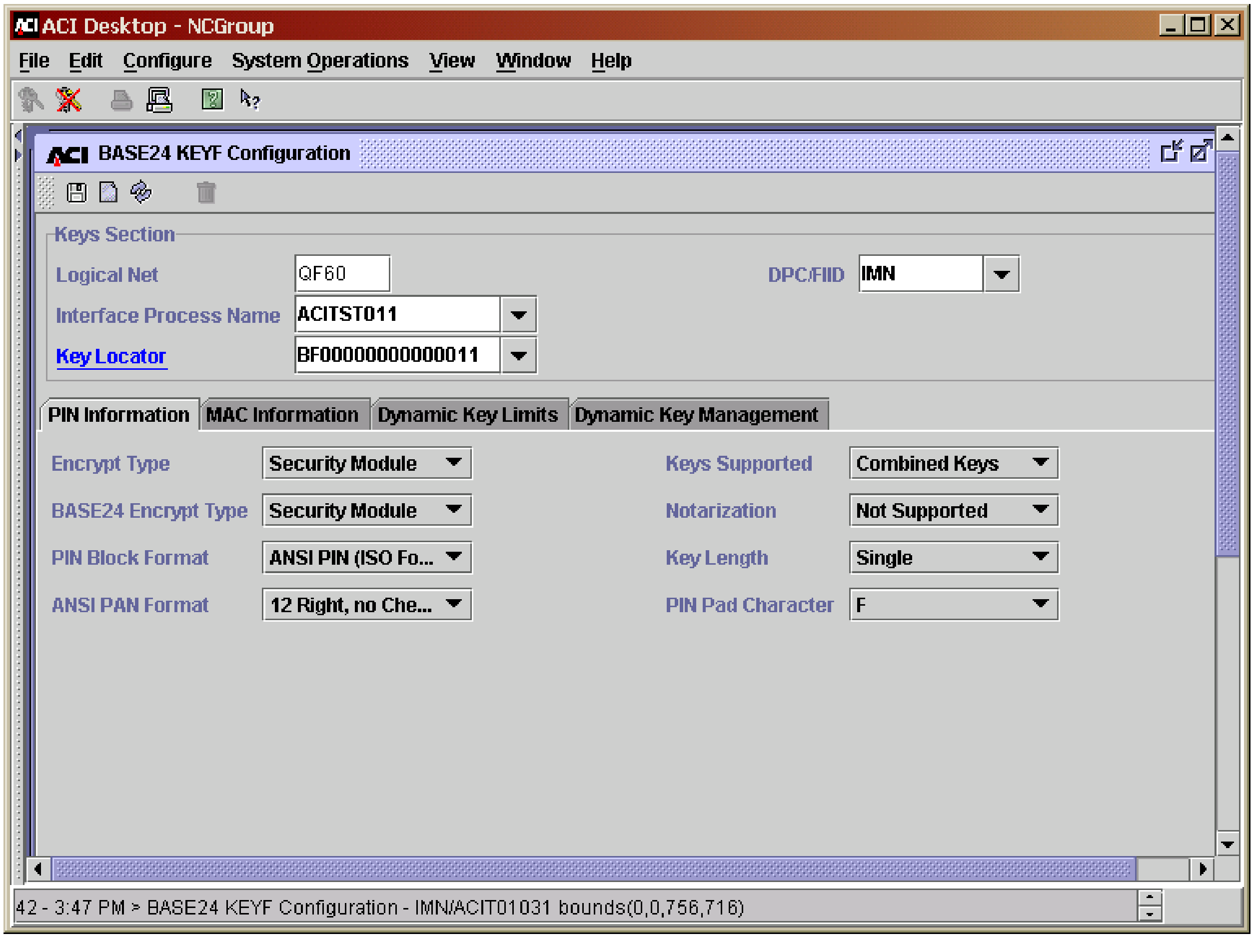
The KEYF conversion program can be run to create the ISECD records while updating the KEYF records with key locators that point to the appropriate ISECD records. The Canadian KEYF conversion program is provided in the CD60UC04 subvolume and it is run by using an obey file (FILKEYFR). For detailed procedures on running this conversion program, please refer to the BASE24 Release 6.0 Canadian Customers Conversion/Implementation Guide.

Once the existing KEYF file is converted, all security data maintenance is performed through the ACI Desktop user interface.

Below is an example when the local node is an **acquirer** (the remote node is an issuer).



Below is an example when the local node is an **issuer** (the remote node is an acquirer).



9.3 Interchange Security File (ISECD)

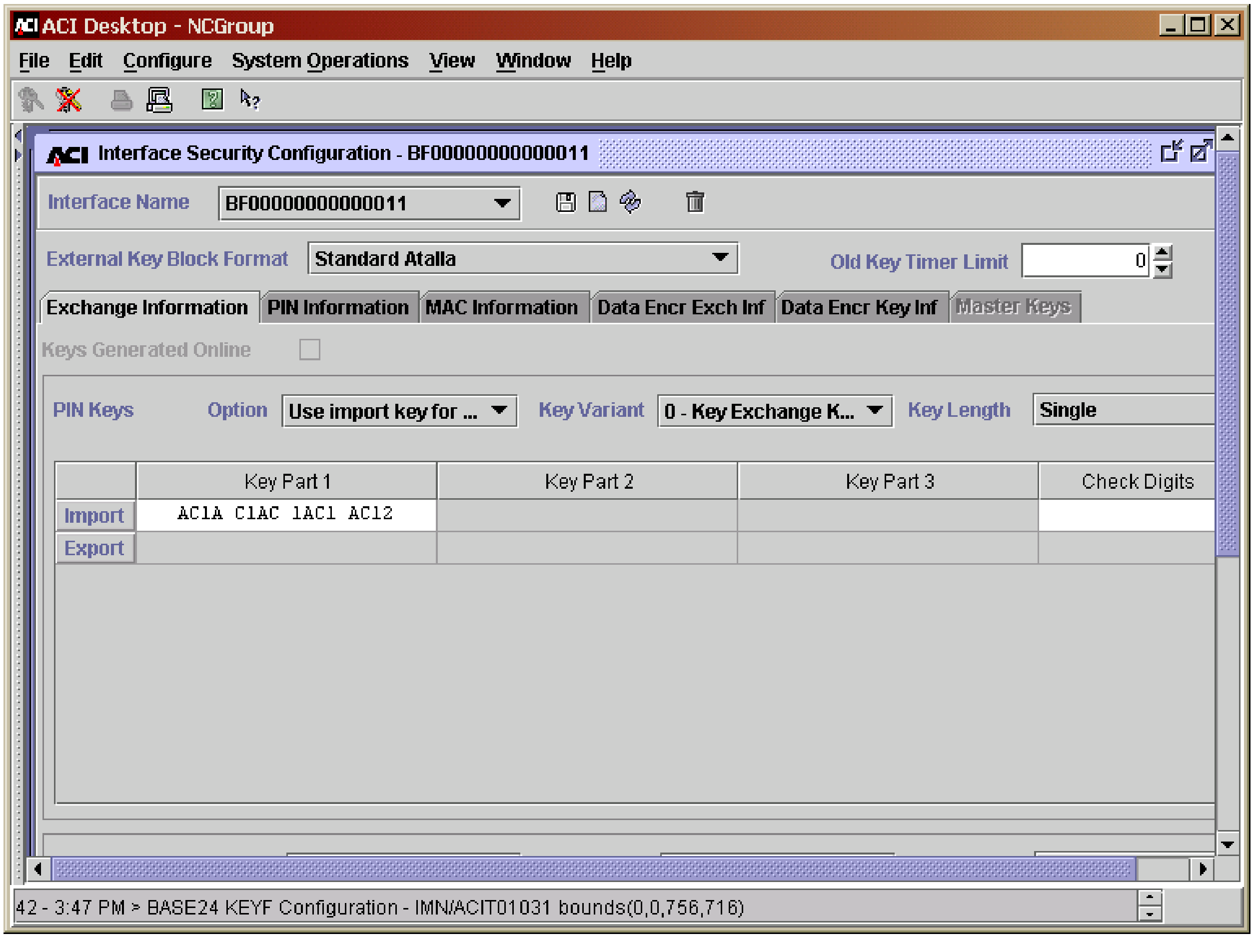
A new database, ISECD, is required. It contains the encryption/decryption keys referenced by key locators and the parameters required by the IMNI/IMNSC process for PIN encryption, PIN translation, message authentication, and data encryption. This file will be created when you run the KEYF conversion program or records can be added, deleted or modified manually from the ACI Desktop.

To support data encryption for the IMNI Interface process, you must configure the data encryption keys and data encryption key exchange keys on the Interface Security Configuration window of the ACI Desktop user interface. This record must correspond to an existing record in the KEYF. If a record does not already exist in the KEYF for the interface, you must first add the record in the BASE24 KEYF Configuration window using the ACI Desktop user interface.

The Interface Security Configuration window has a number of tabs. Depending on the security device used in your environment; the key fields are displayed with the short key block format (Thales e-Security and Atalla non-AKB versions) or the expanded key block format (Atalla AKB versions). Below is an example of each tab on the ISECD screenwhen the local node is an **issuer** (the remote node is an acquirer).

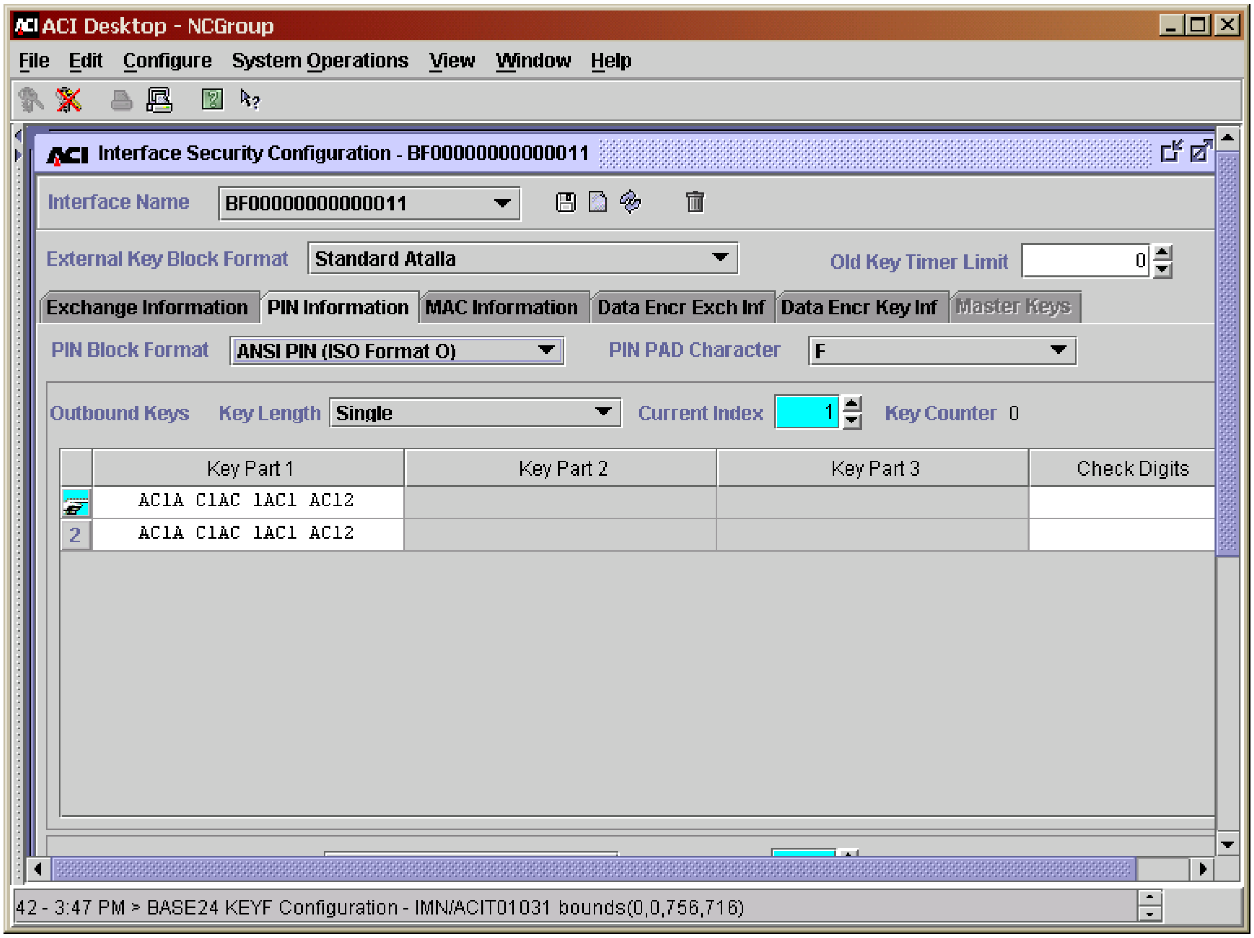
9.3.1 Exchange Information Tab

On the Exchange Information tab, key exchange keys for importing and exporting PIN and MAC keys are defined.
Due to Interac requirements change Old Key Timer Limit to Zero.



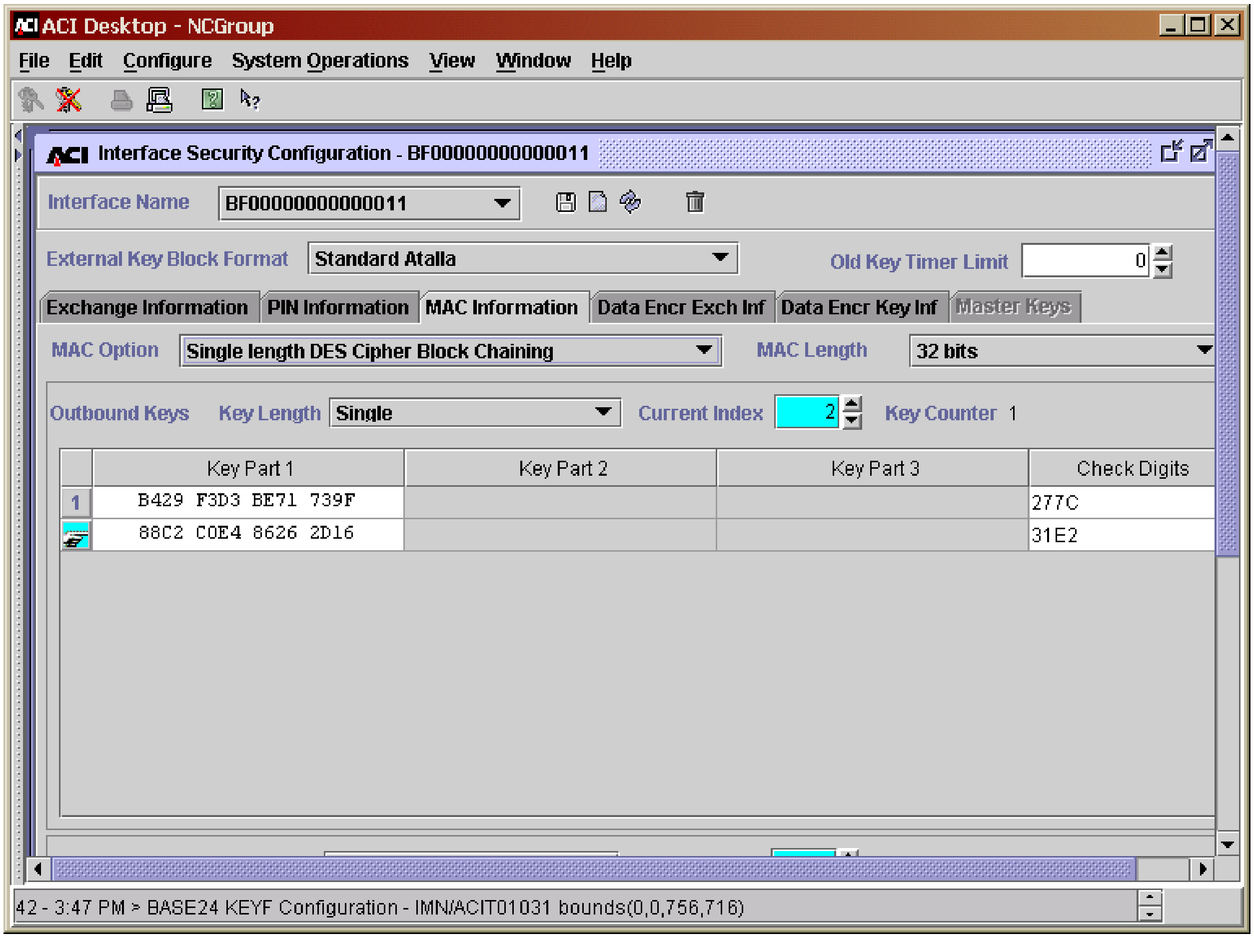
9.3.2 PIN Information Tab

On the PIN Information tab, the current PIN key, the key length of the PIN, the PIN Pad character, and the PIN keys used for outbound requests to an interchange and for inbound requests from an interchange are defined.



9.3.3 MAC Information Tab

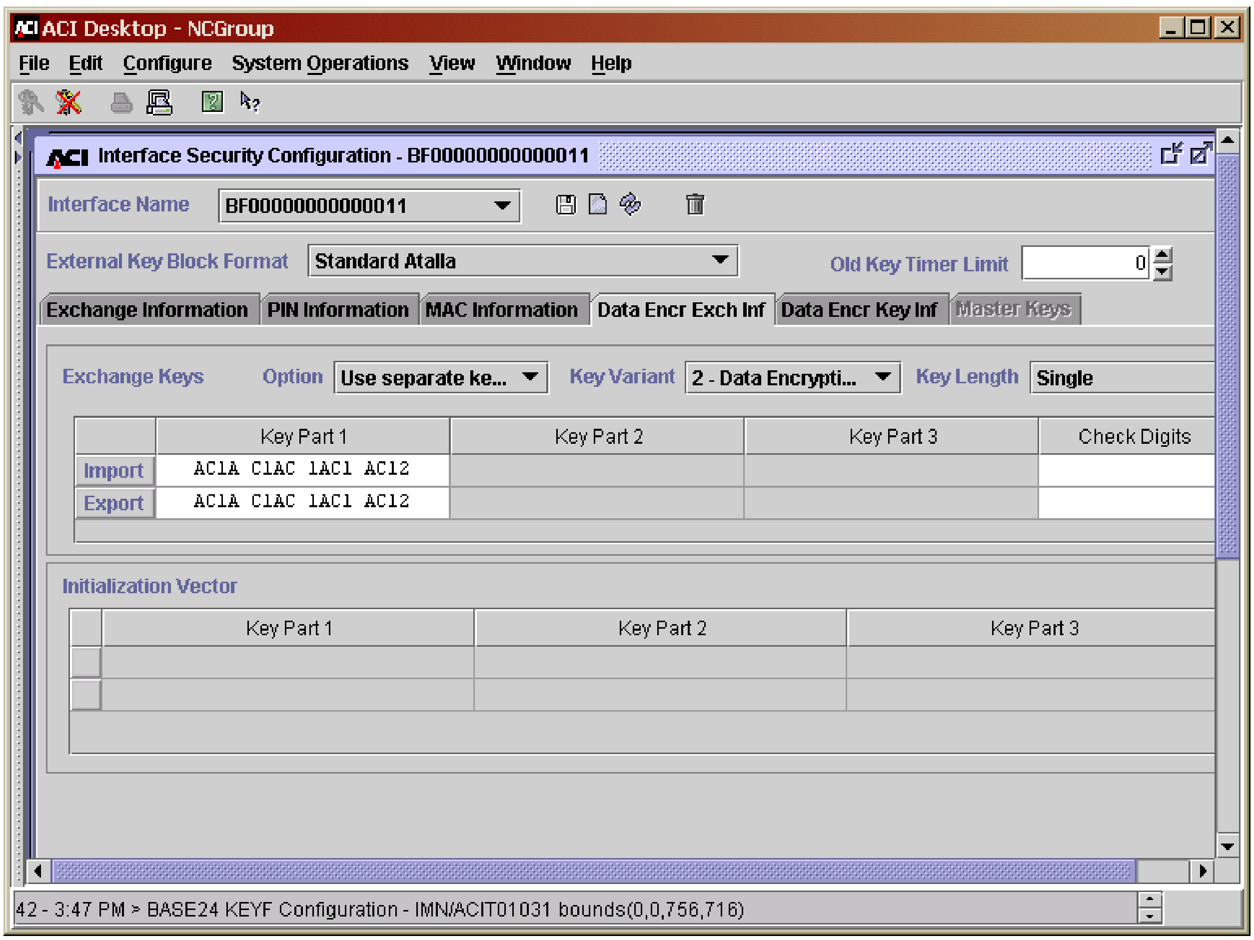
On the MAC Information tab, the current MAC key, the length of the MAC key, the outbound MAC keys used for MAC generation and the inbound MAC keys used for MAC verification are defined.



9.3.4 Data Encr Exch Intf Tab

On the Data Encr Exch Inf tab (for Atalla security devices only), the exchange keys required for data encryption are defined.

Configure the Exchange Keys fields on the Data Encr Exch Inf tab for the key exchange keys to import or export a new data encryption key. The import key is used to decrypt a new data encryption key received from an interface. The export key is used to encrypt a new data encryption key transmitted to an interface. If desired, the import key can be used for both operations. Note that for the IMNI, the Data Encrypt Exchange Key MUST be the same as the PIN and MAC Exchange keys for the 6.0 version of the IMNI to work with a 5.3 version of the IMNI on the remote node.



9.3.5 Data Encr Key Intf Tab

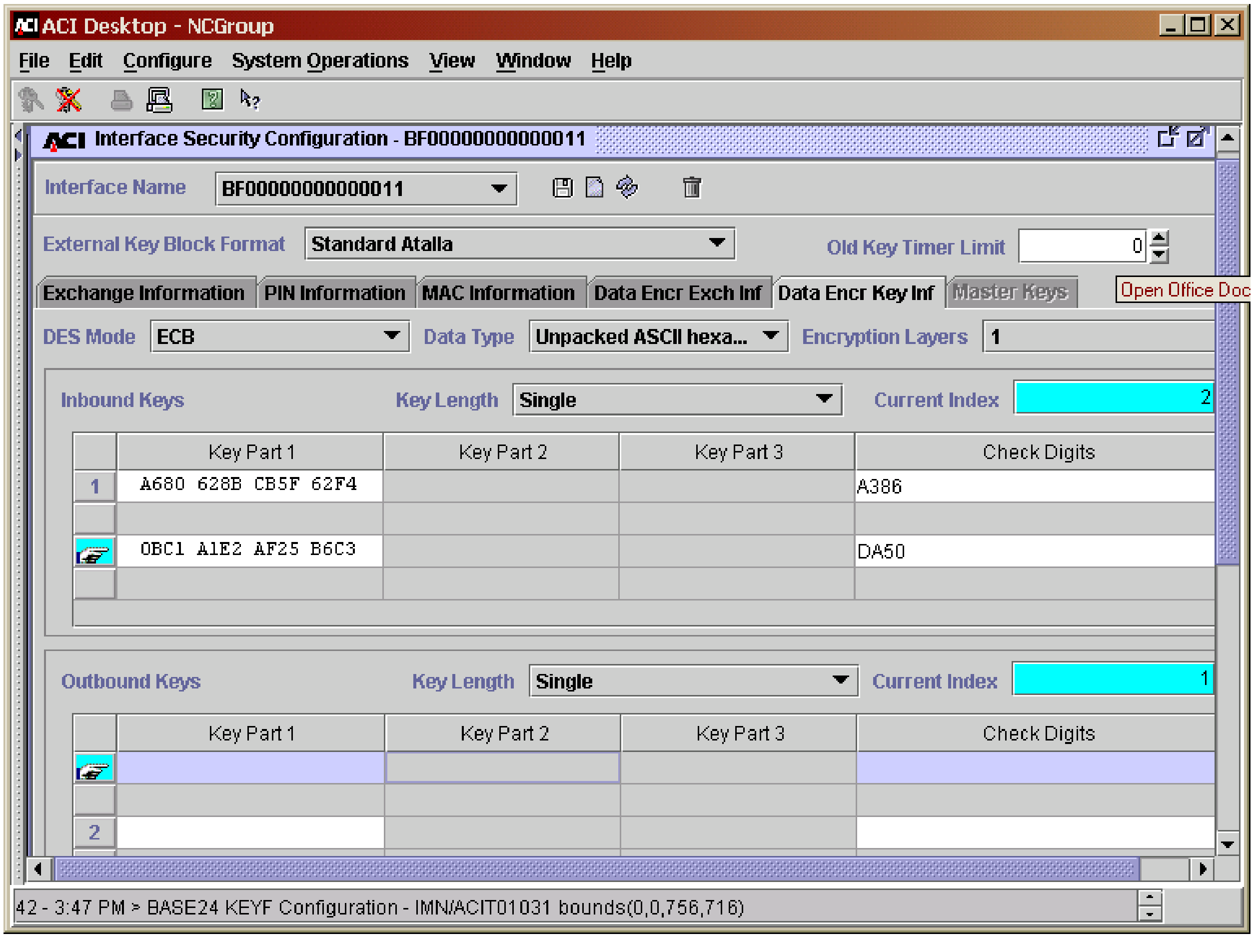
On the Data Encr Key Inf tab (for Atalla security devices only), the type of data encryption, and the data encryption keys used for outbound requests to an interchange and for inbound requests from an interchange are defined. Data can be encrypted using Cipher Block Chaining (CBC) or Electronic Code Book (ECB).

The following fields on the Data Encr Key Intf tab must be configured for the inbound and outbound data encryption keys. The inbound key is used to decrypt or translate encrypted data received from the interface. The outbound key is used to encrypt or translate data to be sent to the interface.

- Inbound Keys (contains one or two data encryption keys, depending on whether you are using one or two layers of data encryption). These keys are used to decrypt or translate encrypted data received from the interface.
- Outbound Keys (contains one or two data encryption keys, depending on whether you are using one or two layers of data encryption). These keys are used to encrypt or translate encrypted data transmitted to the interface.
- DES Encr Mode (specifies whether to use the ECB or CBC mode of DES).
- Data Type (specifies whether you are using unpacked ASCII hexadecimal or binary data encryption).
- Encr Layers (specifies whether to use one or two layers of data encryption).

All of the above information is entered on the Interface Security Configuration window and is stored in the Interface Security File (ISECD).

An example of the Data Encr Key Intf tabs on the Interface Security Configuration window is shown below.



9.3.6 Master Keys Tab

This tab is not applicable for IMNI/IMNSC process.

9.3.7 Interac Terminal Setup

The two fields on screen 6 of the PTD must be updated to state that in order for a terminal to be an Interac terminal, the customer must set Enhanced Preauth to "1", and Preauth Chargebacks to "Y". This combination of settings will tell the IMNI that the terminal is a valid Interac Terminal. More information on Preauth, read Base24-pos Transaction Processing Control.

```
 DEFAULT TRAN CODE: 10  (NORMAL PURCHASE)

ADJUST AMT2 > AMT1: Y (ALLOWED)
      SALES DRAFTS: N (NOT ALLOWED)
      CLERK TOTALS: N (NOT ALLOWED)
   AUTH LESSER AMT: N (NOT ALLOWED)
  ENHANCED PREAUTH: 1 (DEBIT ONLY)
 PREAUTH CHARGEBACKS: Y (SUPPORTED)
```

9.4 CPF/IDF File Setup

9.4.1 CPF Setup (Example)

The Card Prefix setup for routing of cards belonging to an institution must be setup using CPF/IDF, and the IMN BIN Table must contain the prefix.

```
BASE24-BASE    CARD PREFIX            POS8  0297  07/11/27  10:39  01 OF 22
 PREFIX: 8888                 PAN LENGTH: 16                FIID: 0297


                           CARD TRACK INFORMATION
    CARD TYPE: P   (PROP DEBIT  )          CARD PROFILE:        MBR LENGTH: 1
TRACK 1 SETTINGS:                  TRACK 2 SETTINGS:       TRACK PREFERENCE: 0
      MBR #:   0 POFST/PVV:   0          MBR #:   0 POFST/PVV:  26
ALGO #/PVKI:   0  EXP DATE:   0    ALGO #/PVKI:   0  EXP DATE:  18
LENGTH MIN/MAX:  0 /  0            LENGTH MIN/MAX:  0 /  0      BAD TRK LEN: 0

                           PROCESSING INFORMATION
       PAN ACCESS TYPE: 0   (MBR 0)              PREFIX ROUTING: 6
        EXP CHECK TYPE: 0   (NO CHECK)              MOD10 CHECK: 0 (NO CHECK   )
                         CARD PROCESSING INFORMATION
        ACTIVITY LIMITS:      TOTAL           OFFLINE
               CASH WDL:             9999             9999
               CASH ADV:             9999             9999
                   AGGR:            99999            99999
 RECORD LAST CHANGED: 07/11/26  14:33   BY USER: 0255 , 00000255   ADD
 *********************************** BASE24 ***********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                     F12-HELP
```

9.4.2 IDF Setup (Example)

The Institution definition file setup for routing of cards belonging to an institution must be setup using CPF/IDF, and the IMN BIN Table must contain the prefix

```
BASE24-BASE  INSTITUTION FILE         POS8  0297  07/11/27  10:54  01 OF 42

                  FIID: 0297        FI-NAME: IMN INSTITUTION

  STATE:   0   COUNTY:    0   COUNTRY:  124   PHONE:
INSTITUTION ID NUMBER:  00000700297   REFRESH GROUP:


                                  FILE NAMES:

                  NEG:
                  UAF:
                  CAF:
                  SPF:
                 PBF1:
                 PBF2:
                 PBF3:
                 PBF4:

 RECORD LAST CHANGED: 07/11/26  14:33   BY USER: 0255 , 00000255   CHANGE
 *********************************** BASE24 ***********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                       F12-HELP
```

```
BASE24-POS   INSTITUTION FILE         POS8  0297  07/11/27  10:55  16 OF 42

                  FIID: 0297        FI-NAME: IMN INSTITUTION

                           POS ROUTING TABLE

PRIMARY     SYMBOLIC     ACCT  PRFX  AUTH          AUTH           HOLDS
  DPC         NAME       TYPE   RTG  TYPE (DESCR)   LVL (DESCR)    LVL (DESCR)
  297   P8P^IMNI          AL     A     0 (HOST)      1 (ONLINE)     0 (N/A)








 CHF FILE NAME:                                          LOG ROUTING CODE: 0002
 RECORD LAST CHANGED: 07/11/26  14:33   BY USER: 0255 , 00000255   CHANGE
 *********************************** BASE24 ***********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                       F12-HELP
```

APPENDIX A: ILF SUBSTATE TABLE

The following information describes the reason of authorization when an ILF record is logged with a particular ILF substate.

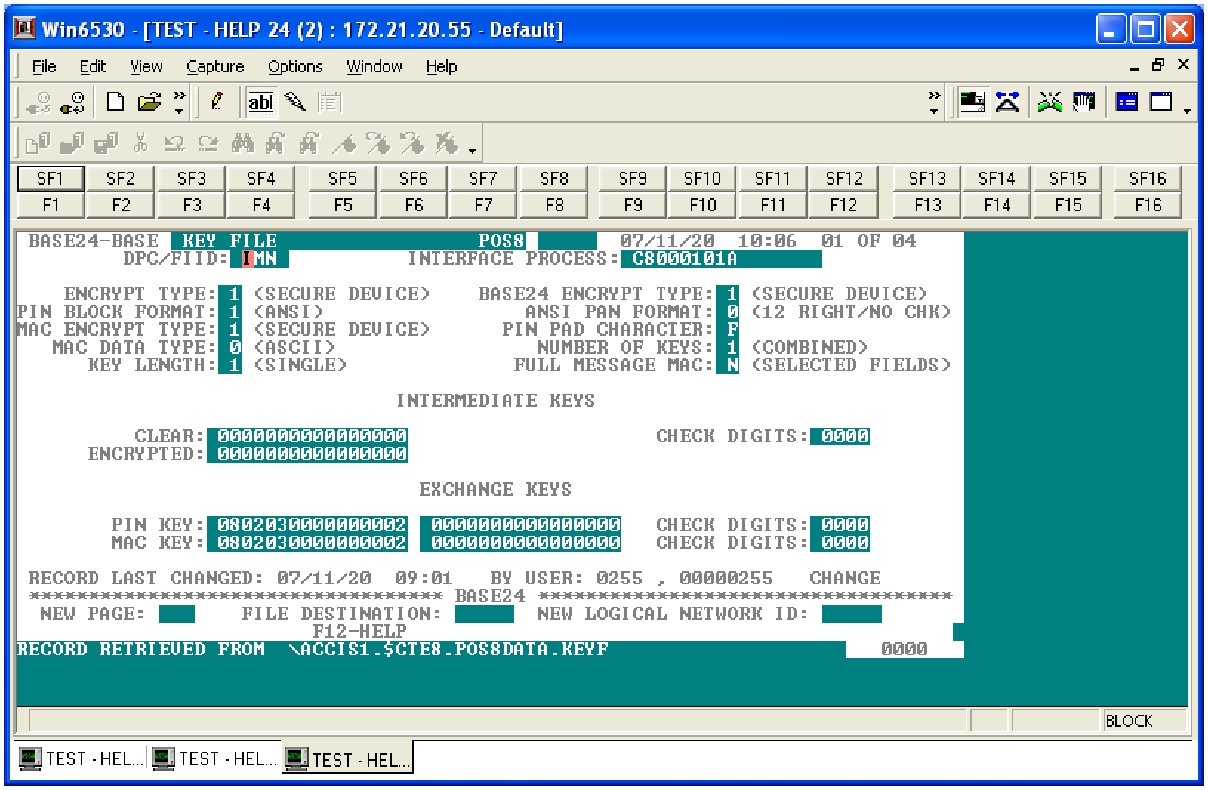
Substate Literal Name	Substate Value	Description
SUB-OK-L	0	No error occurred when a transaction is processed.
SUB-SYSTEM-ERROR-L	1	Error on the transmission of a SEM (Standard External Message).
SUB-FRMT-ERR-L	2	Error formatting a PSTM into a SEM, the transaction will not be forwarded.
SUB-FAIL-PRE-AUTH-L	3	Not used.
SUB-RQST-DOWN-L	4	Failed on sending the response message to the BASE24 authorization process.
SUB-TIMEOUT-L	5	Outbound transaction time out.
SUB-RESP-DOWN-L	6	Failed on attempt to resend the request message on another Entry Point or the issuer or when the acquirer receives the SAF response with the following IMN status code: "80" – Undeliverable "87" – Pacing connection blocked "91" – AIL not available
SUB-RVSL-NOT-FOUND-L	7	Not used.
SUB-MISC-L	8	Not used.
SUB-MAC-FAIL-L	11	No keys are available.
SUB-SAF-SEM-ONLY-L	12	Only the SEM portion was available at the time when the transaction was logged. This substate occurs when an IMNI/IMNSC switches its SAF transactions to another IMNI/IMNSC process.

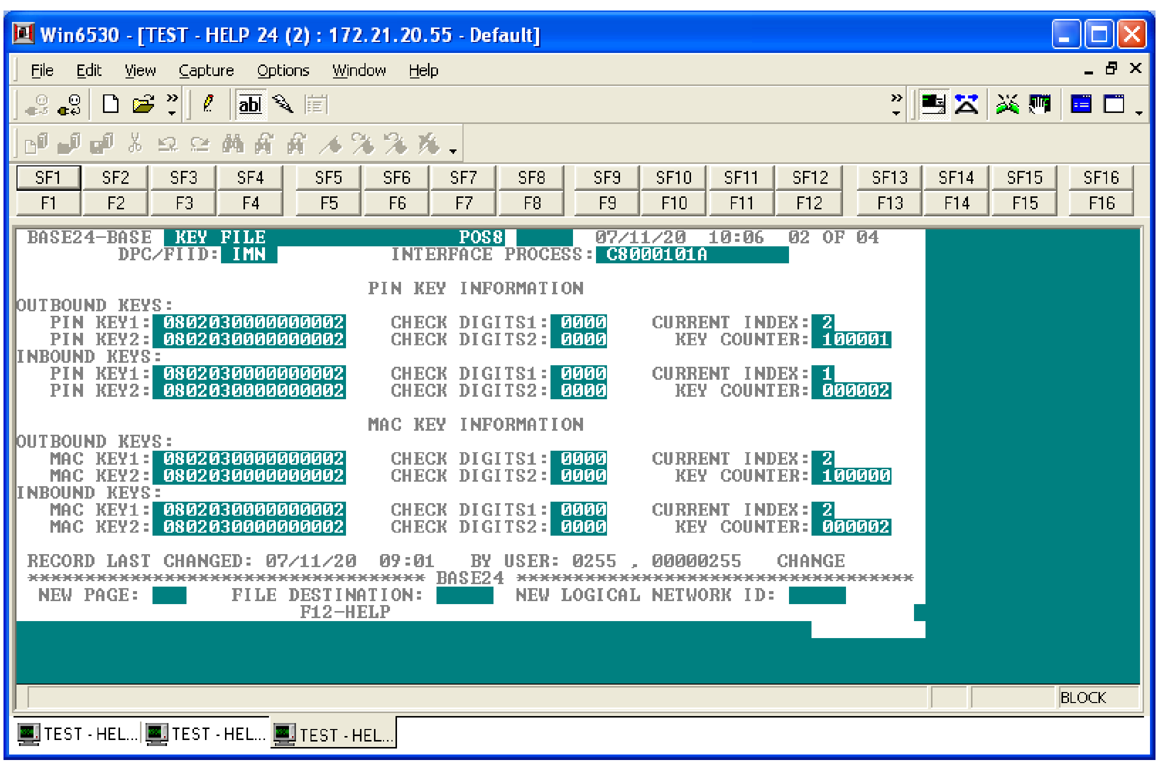
Substate Literal Name	Substate Value	Description
SUB-SAF-DELETED-ERROR-L	13	SAF record has been deleted due to a SAF response received with one of the following response code: "6" - Edit check error "12" - Invalid processing code "13" - Invalid amount "81" - Invalid PIN block "82" - PIN length error "92" - Financial institution or intermediate network facility cannot be found for routing "Q0" - Capture date error
SUB-SAF-DELETED-CUTOVER-L	14	SAF 420's deleted at cutover time with reversal reason code of timeout.
SUB-SAF-REQUEST-L	15	SAF request message is generated for message type 220, 402, 420, 6xx (Admin).
SUB-SAF-DENIED-L	16	SAF request is denied.
SUB-ILF-NORMAL-L	17	Transaction response received and successfully sent by issuer.
SUB-SAF-UNMATCHED-RESP-L	18	Cannot find original SAF request for the response.
SUB-SAF-RETRY-MAX-L	19	Not used.
SUB-LATE-RESP-L	20	Received late response from IMN or BASE24.
SUB-SECURITY-ERROR-L	21	MAC generation or verify failure as well as any security error such as invalid security block or invalid key or key synchronized error.
SUB-SAF-UNACKNLDG-RVSL-L	22	SAF response with invalid capture date – SAF deleted.
SUB-SAF-DELETED-RESP-OK-L	23	SAF response message received OK. Request message deleted from SAF.
SUB-SAF-DELETED-CMD-L	24	SAF request deleted from an operator's command.
SUB-INVALID-DESTN-L	30	Destination node ID not found. Checked on acquirer side.
SUB-INVALID-ROUTE-L	31	PAN routing not found in BIN table. Checked on acquirer side.
SUB-POS-INVALID-L (IMNI) SUB-ATM-INVALID-L (IMNSC)	32	Transaction not processed. Transaction type not entered on ICF list of allowed transactions. Checked on acquirer side.

Substate Literal Name	Substate Value	Description
SUB-SWITCH-ORIG-L	33	Switch originated transaction possible duplicated.
SUB-NOT-INTERAC-L	34	Not an Interac transaction.
SUB-INVALID-TRANS-L	35	Transaction type invalid.
SUB-INVALID-SERVICE-L	36	Service not allowed.
SUB-UNABLE-TO-SEND-L	37	Unable to send transaction because destination is in forward mode only.
SUB-INVALID-AMT-L	38	The amount is too large for switch variables.
SUB-ALT-ROUTE-L	39	Unable to alternate route transaction.
SUB-INVALID-DATE-L	40	Invalid transaction date.
SUB-INTERNAL-ERROR-L	41	Not used.
SUB-NOT-PROCESSED-L	42	Transaction not processed due to error in SEM is found.
SUB-CAPDATE-CHANGED-L	43	Capture date is changed at cutover period.

APPENDIX B: TSS SETUP - KEYF TABLE

KEYF – BASE24



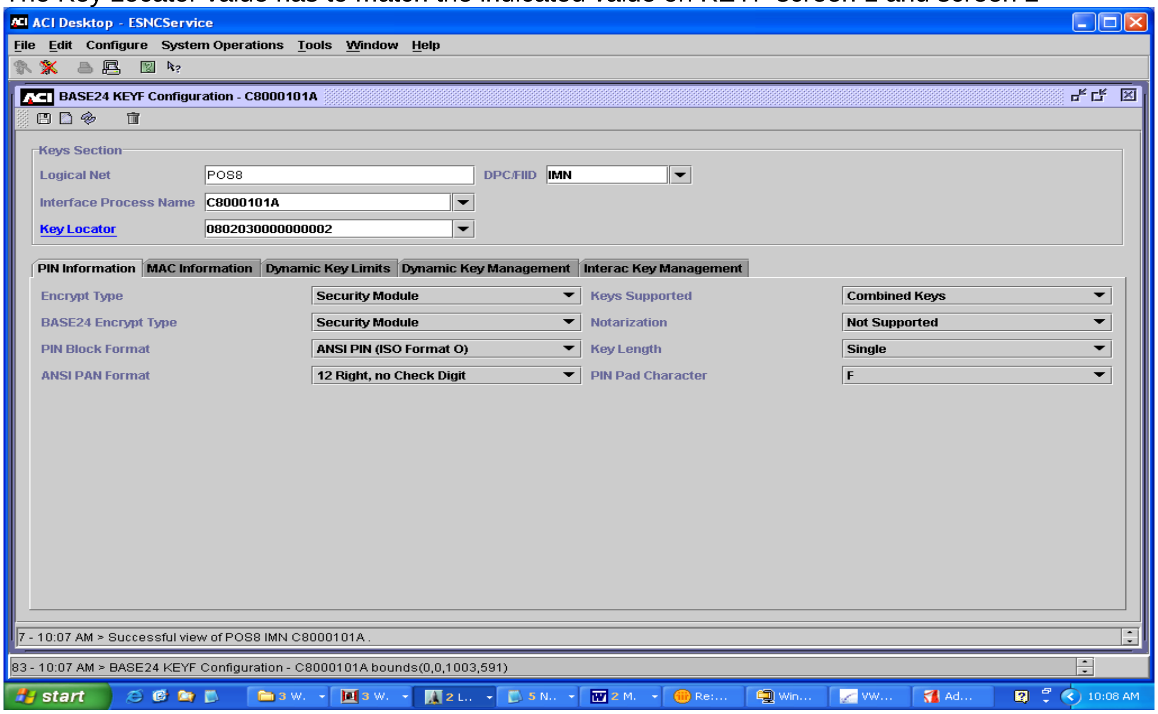


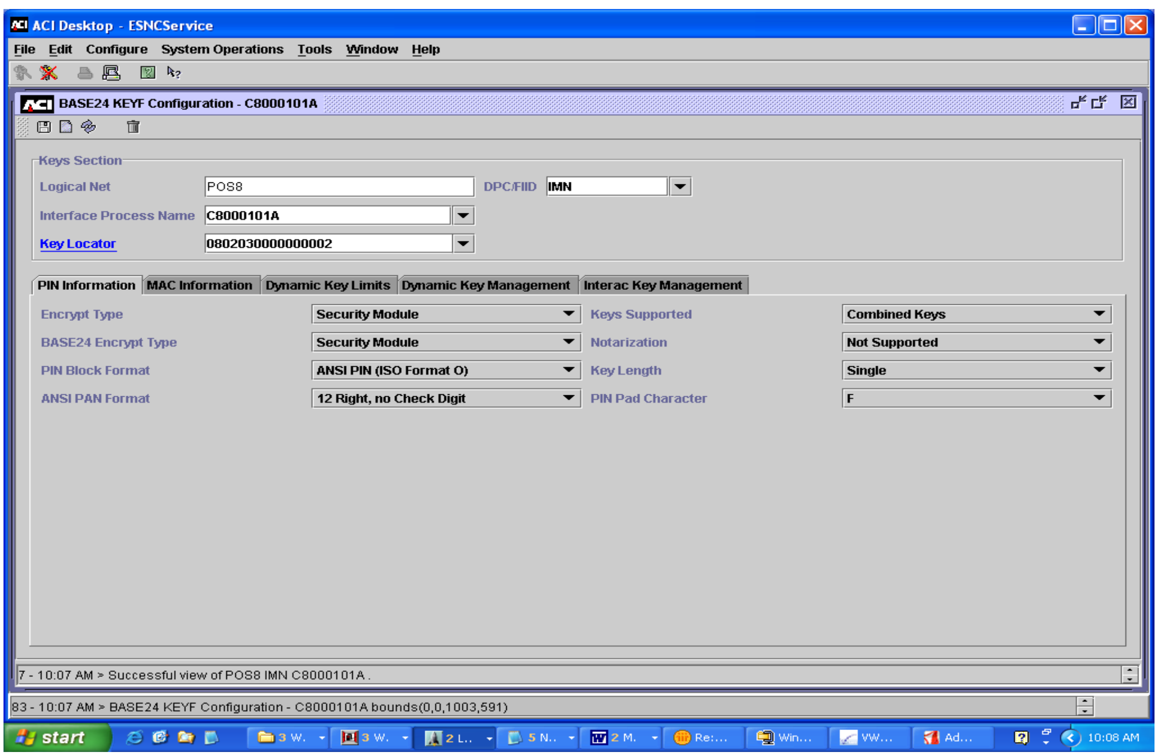
TSS SETUP

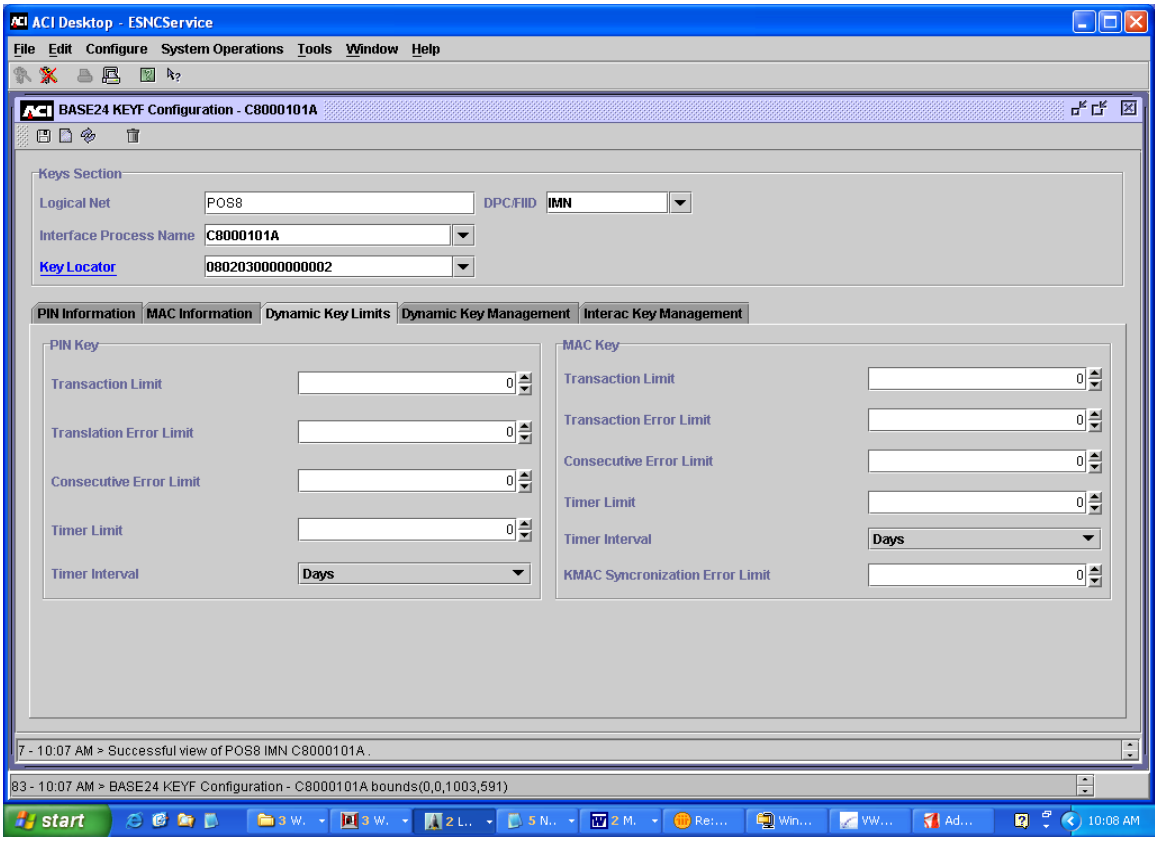
BASE24 KEYF CONFIGURATION

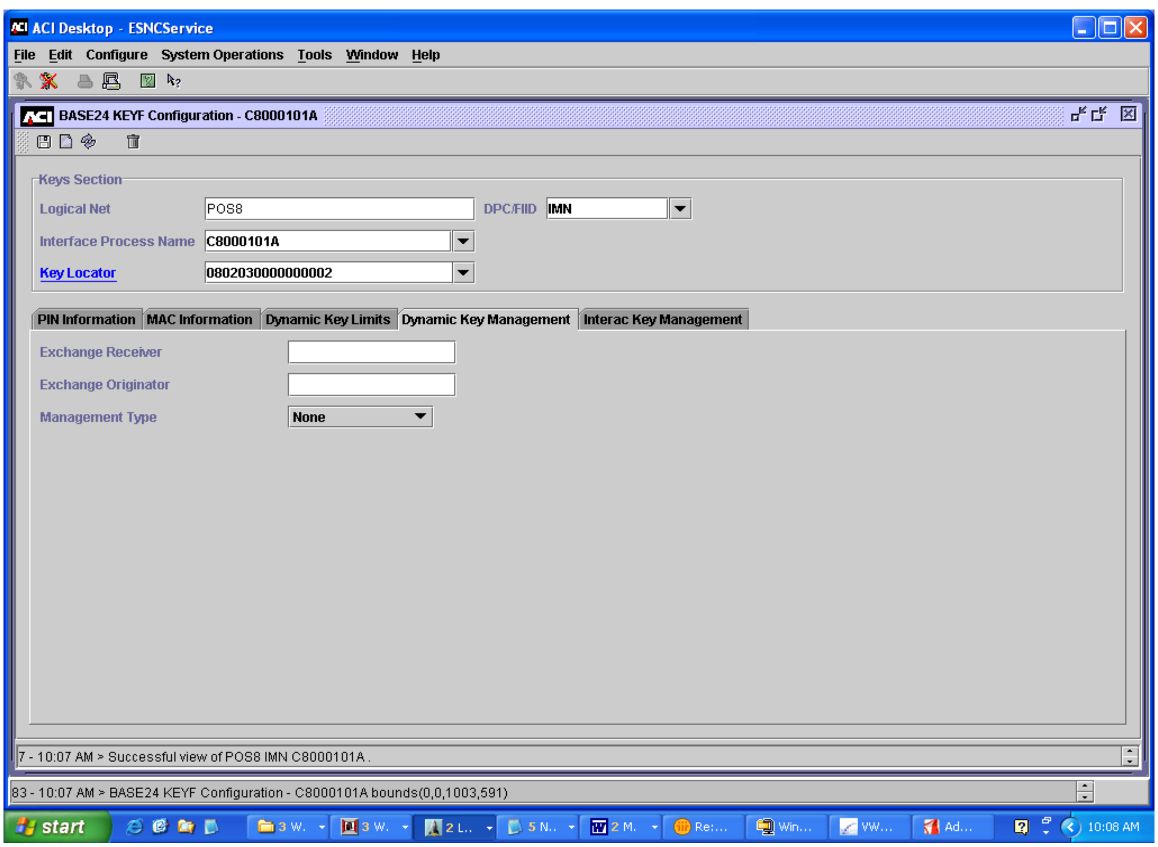
Configure > Transaction Security >BASE24 KEYF Configuration

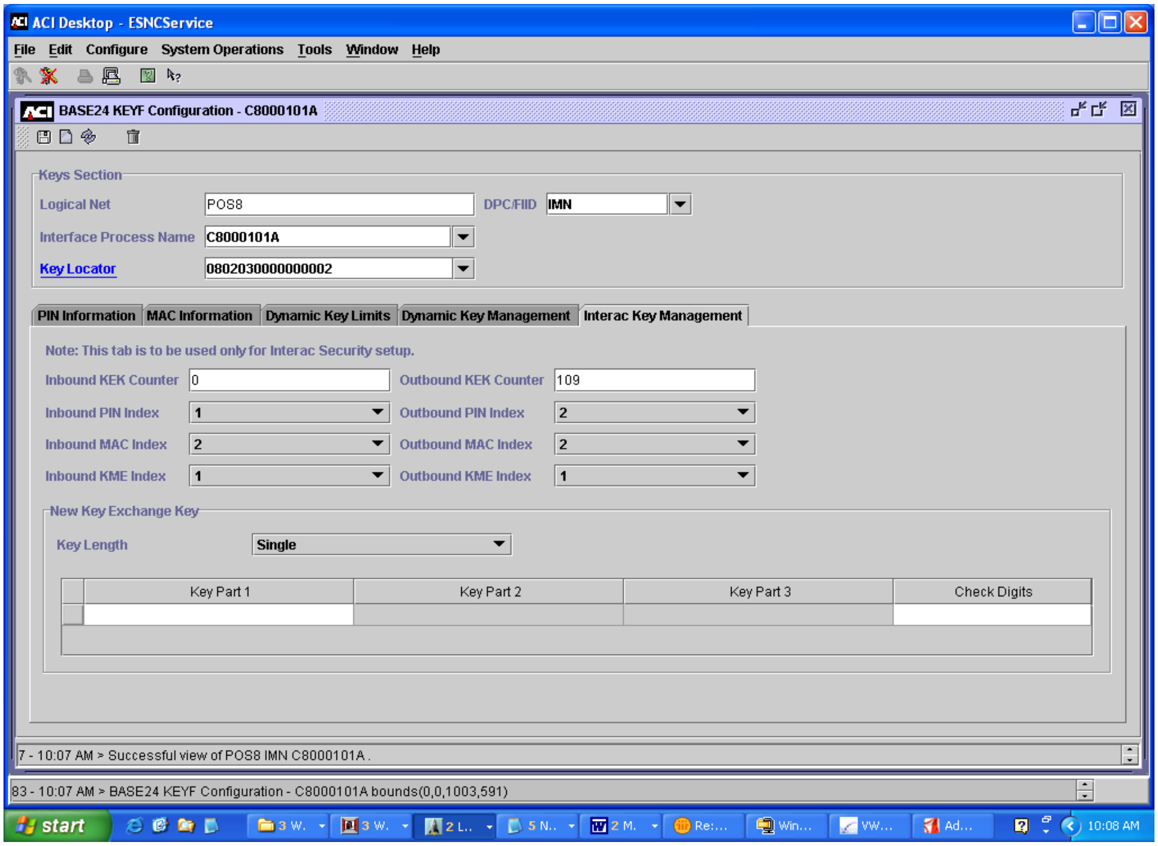
The Key Locator value has to match the indicated value on KEYF screen 1 and screen 2






ACI Desktop - ESNCService
File Edit Configure System Operations Tools Window Help
BASE24 KEYF Configuration - C8000101A
Keys Section
Logical Net
POS8
DPC/FIID
IMN
Interface Process Name
C8000101A
Key Locator
0802030000000002
PIN Information
MAC Information
Dynamic Key Limits
Dynamic Key Management
Interac Key Management
PIN Key
Transaction Limit
Translation Error Limit
Consecutive Error Limit
Timer Limit
Timer Interval
Days
MAC Key
Transaction Limit
Transaction Error Limit
Consecutive Error Limit
Timer Limit
Timer Interval
Days
KMAC Syncronization Error Limit
7 - 10:07 AM > Successful view of POS8 IMN C8000101A .
83 - 10:07 AM > BASE24 KEYF Configuration - C8000101A bounds(0,0,1003,591)
start
10:08 AM


ACI Desktop - ESNCService
File Edit Configure System Operations Tools Window Help
BASE24 KEYF Configuration - C8000101A
Keys Section
Logical Net
POS8
DPC/FIID
IMN
Interface Process Name
C8000101A
Key Locator
0802030000000002
PIN Information
MAC Information
Dynamic Key Limits
Dynamic Key Management
Interac Key Management
Exchange Receiver
Exchange Originator
Management Type
None
7 - 10:07 AM > Successful view of POS8 IMN C8000101A .
83 - 10:07 AM > BASE24 KEYF Configuration - C8000101A bounds(0,0,1003,591)
start
10:08 AM

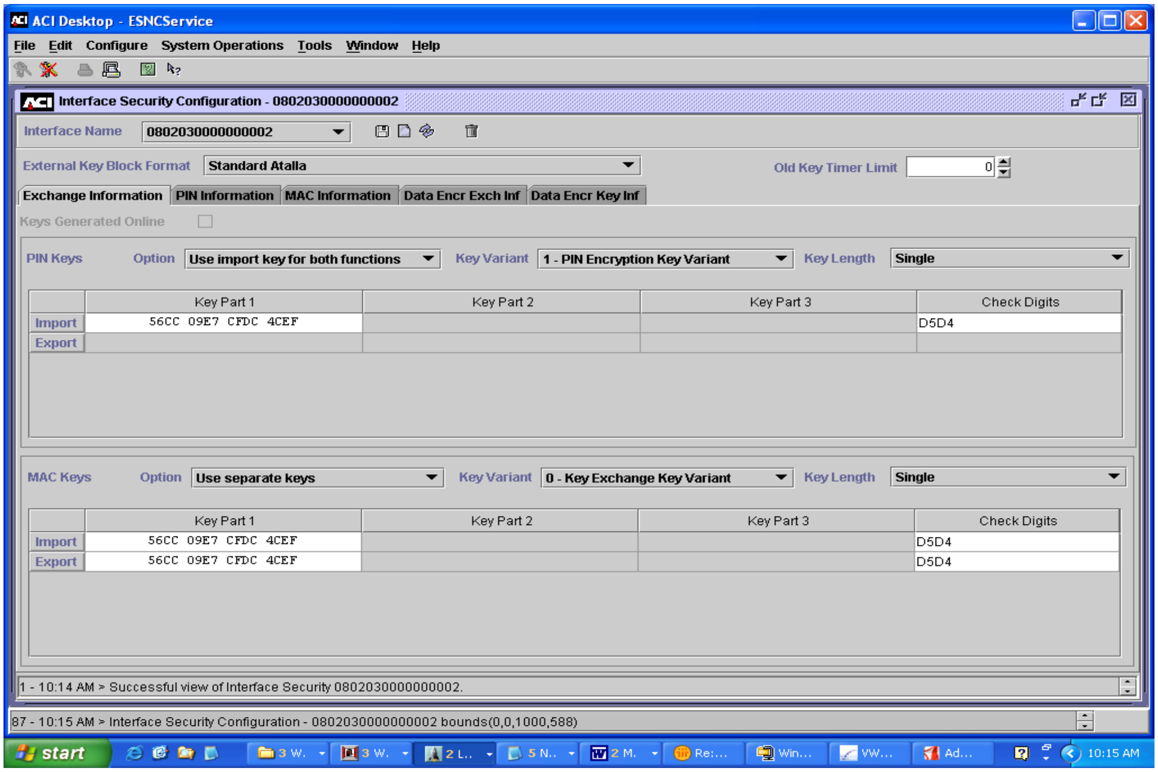


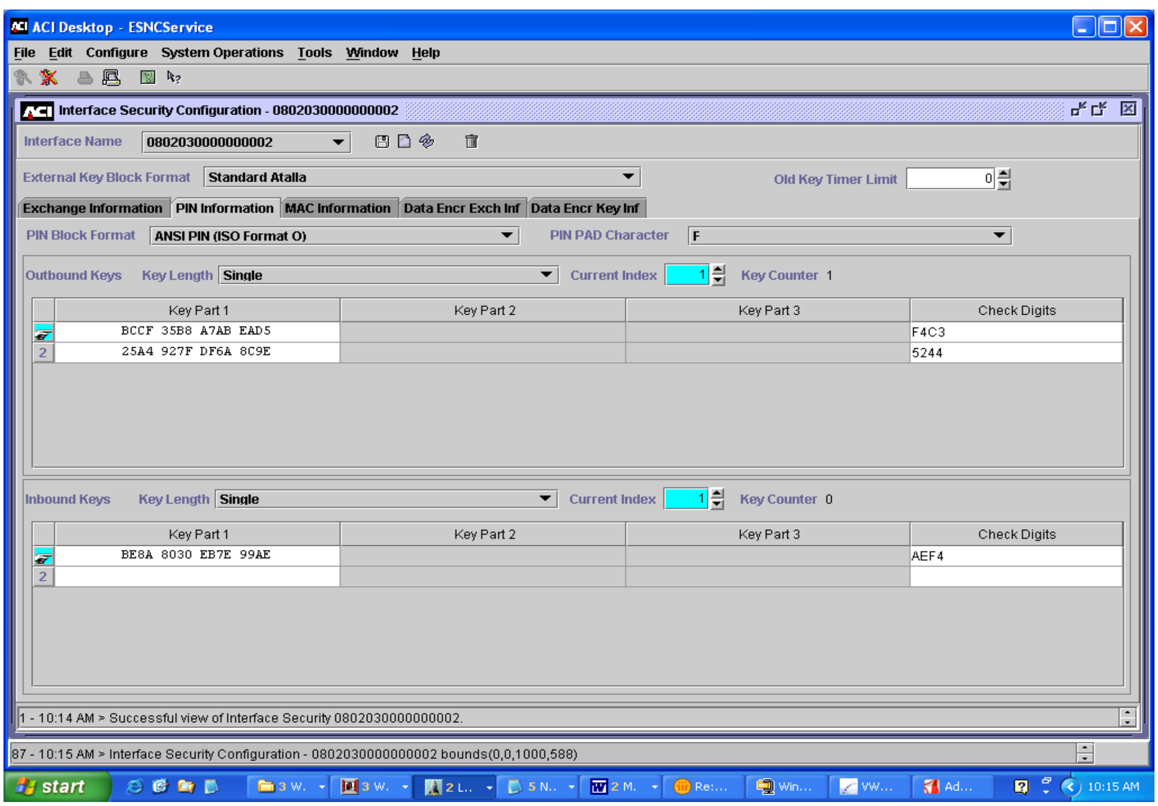
The values of Inbound PIN Index and Outbound PIN Index have to match the values of PIN KEY INFORMATION – CURRENT INDEX (KEYF screen 2)

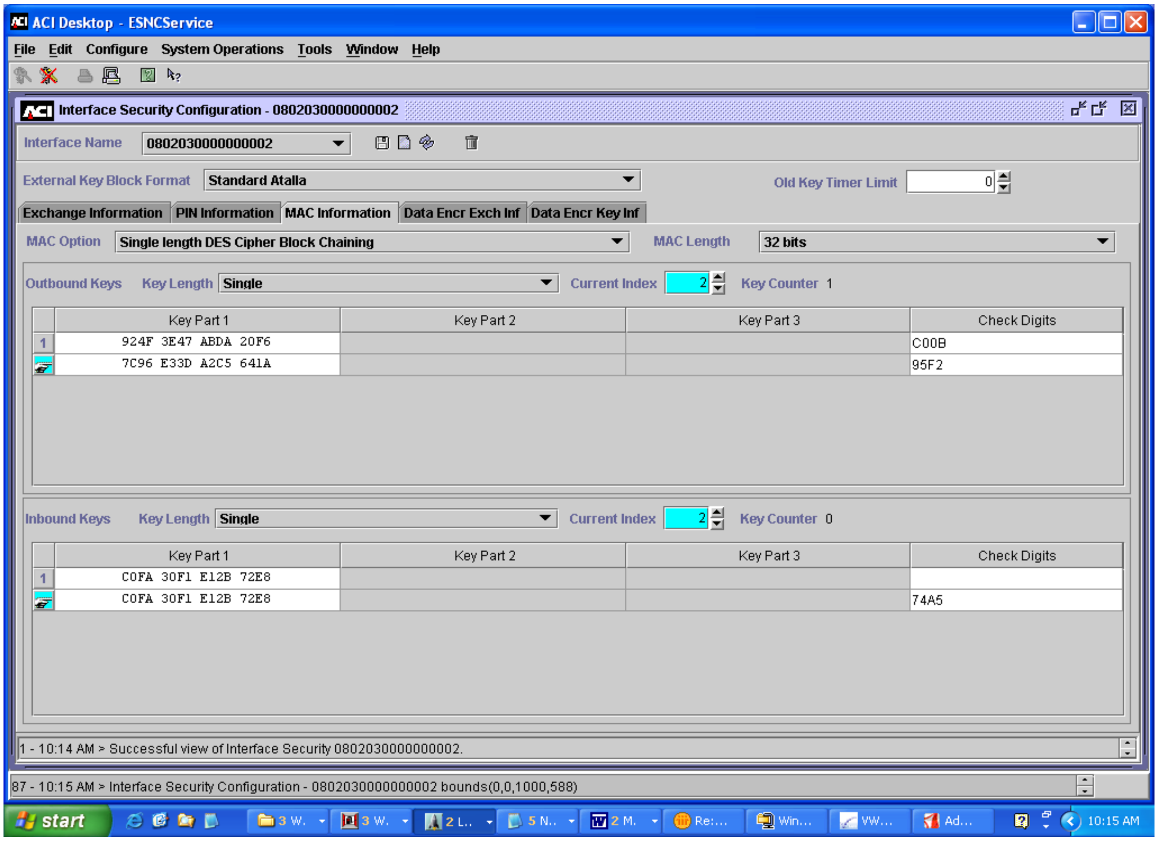
The values of Inbound MAC Index and Outbound MAC Index have to match the values of MAC KEY INFORMATION – CURRENT INDEX (KEYF screen 2)

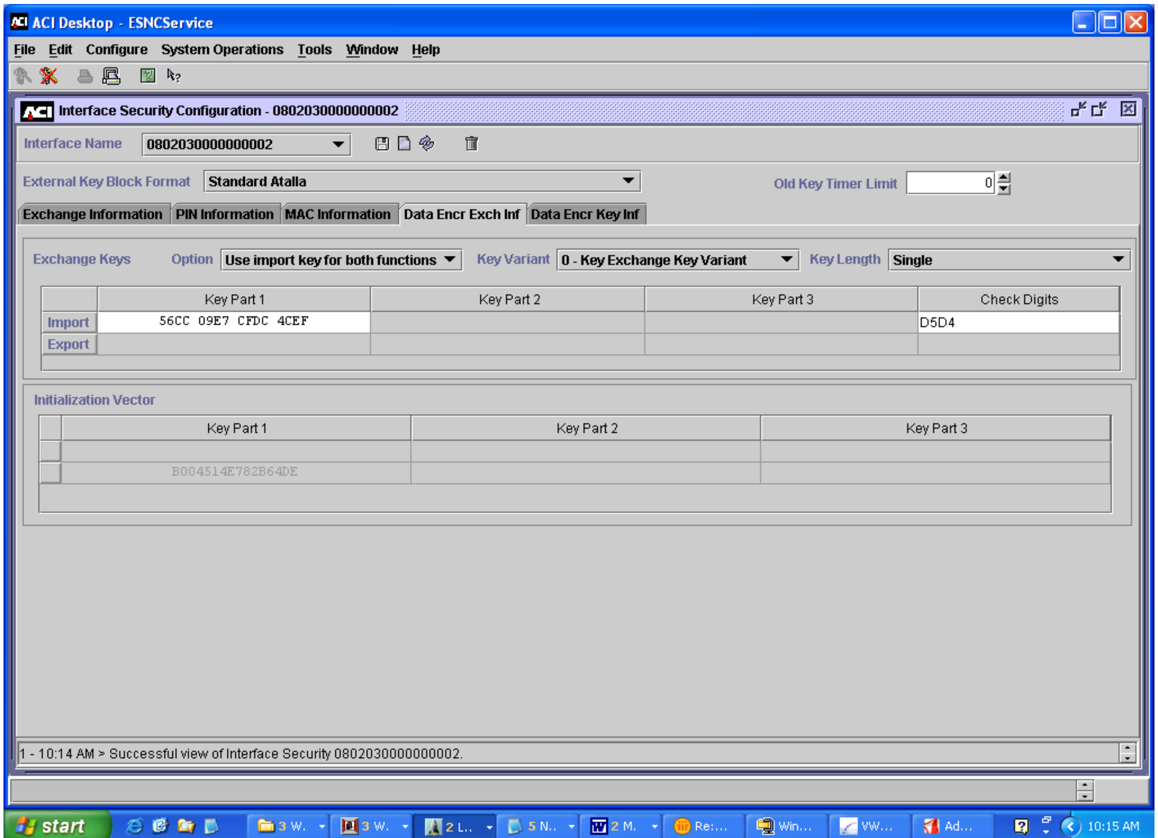
KEY LOCATOR (Interface Security Configuration)

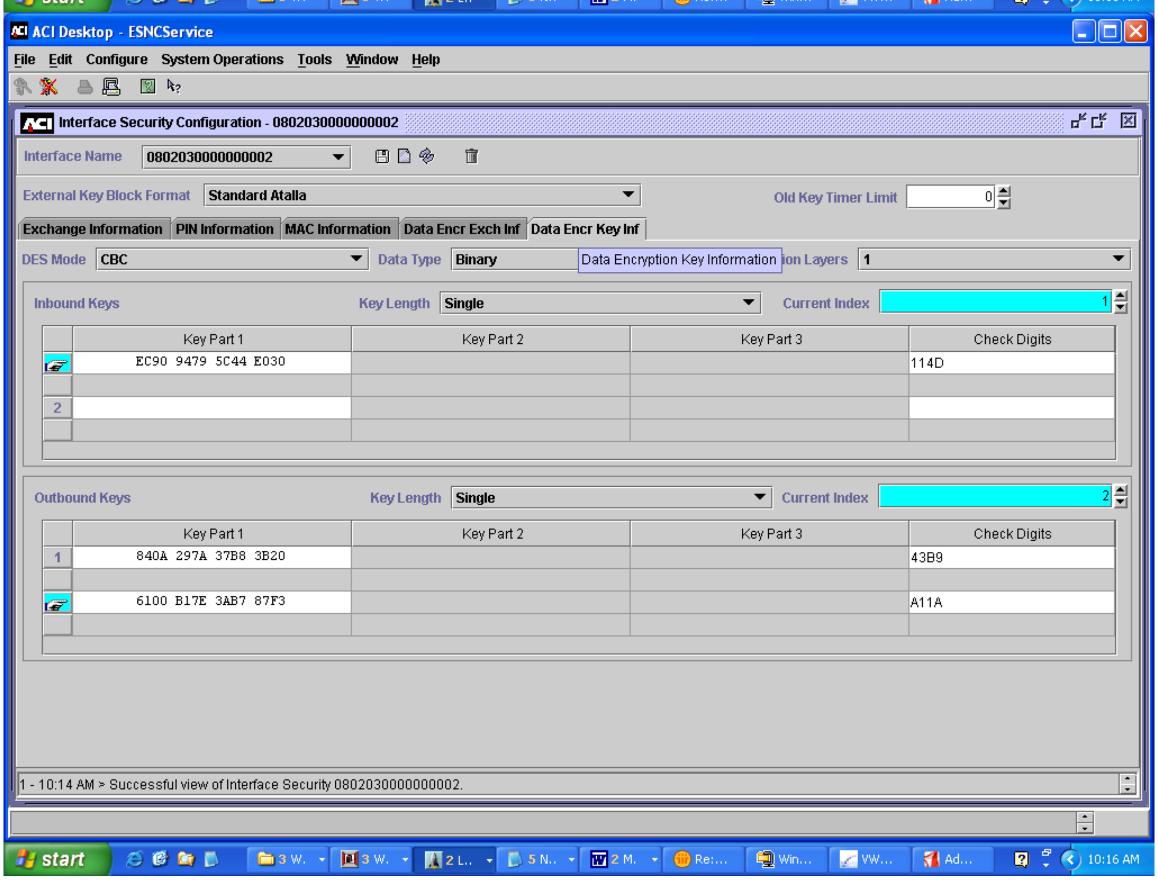
For IMNi and IMNsc set Old Key Timer Limit to zero (0)






ACI Desktop - ESNCService
File Edit Configure System Operations Tools Window Help
Interface Security Configuration - 0802030000000002
Interface Name 0802030000000002
External Key Block Format Standard Atalla
Old Key Timer Limit 0
Exchange Information PIN Information MAC Information Data Encr Exch Inf Data Encr Key Inf
MAC Option Single length DES Cipher Block Chaining
MAC Length 32 bits
Outbound Keys Key Length Single Current Index 2 Key Counter 1
Key Part 1 Key Part 2 Key Part 3 Check Digits
1 924F 3E47 ABDA 20F6 C00B
7C96 E33D A2C5 641A 95F2
Inbound Keys Key Length Single Current Index 2 Key Counter 0
Key Part 1 Key Part 2 Key Part 3 Check Digits
1 C0FA 30F1 E12B 72E8
C0FA 30F1 E12B 72E8 74A5
1 - 10:14 AM > Successful view of Interface Security 0802030000000002.
87 - 10:15 AM > Interface Security Configuration - 0802030000000002 bounds(0,0,1000,588)
start
10:15 AM


ACI Desktop - ESNCService
File Edit Configure System Operations Tools Window Help
Interface Security Configuration - 0802030000000002
Interface Name 0802030000000002
External Key Block Format Standard Atalla
Old Key Timer Limit 0
Exchange Information PIN Information MAC Information Data Encr Exch Inf Data Encr Key Inf
Exchange Keys Option Use import key for both functions Key Variant 0 - Key Exchange Key Variant Key Length Single
Key Part 1 Key Part 2 Key Part 3 Check Digits
Import 56CC 09E7 CFDC 4CEF D5D4
Export
Initialization Vector
Key Part 1 Key Part 2 Key Part 3
B004514E782B64DE
1 - 10:14 AM > Successful view of Interface Security 0802030000000002.
start 10:15 AM


ACI Desktop - ESNCService
File Edit Configure System Operations Tools Window Help
Interface Security Configuration - 0802030000000002
Interface Name 0802030000000002
External Key Block Format Standard Atalla
Old Key Timer Limit 0
Exchange Information PIN Information MAC Information Data Encr Exch Inf Data Encr Key Inf
DES Mode CBC Data Type Binary Data Encryption Key Information ion Layers 1
Inbound Keys Key Length Single Current Index 1
Key Part 1 Key Part 2 Key Part 3 Check Digits
EC90 9479 5C44 E030 114D
2
Outbound Keys Key Length Single Current Index 2
Key Part 1 Key Part 2 Key Part 3 Check Digits
1 840A 297A 37B8 3B20 43B9
6100 B17E 3AB7 87F3 A11A
1 - 10:14 AM > Successful view of Interface Security 0802030000000002.
start 10:16 AM